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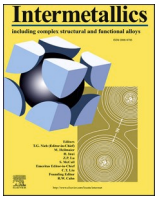

Oxidation behaviors of CrNb, CrNbTi, and CrNbTaTi concentrated refractory alloys

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ABSTRACT

The 1200 °C oxidation behaviors of CrNb, CrNbTi and CrNbTaTi are reported. All alloys formed some combination of BCC and Laves phases. However, additions of Ti produced a unique inflection point where the microstructure inverts to a Ti-rich BCC matrix with Laves phase precipitates. Further additions of Ta to CrNbTi were found to increase the overall phase fraction of BCC phase. The oxidation kinetics for CrNb and CrNbTi were similar and nearly parabolic throughout the duration of exposure. CrNbTaTi exhibited inferior oxidation kinetics and mixed mode behavior throughout oxidation testing. All alloys formed intricate multi-phase oxide structures, consisting of a combination of both simple and complex oxides. The resulting microstructures are characterized using a variety of analytical techniques and the results are discussed relative to modeling predictions using ab initio density functional theory (DFT).

1. Introduction

Recent advancements in aerospace technologies are necessitating the development of materials that can withstand environments beyond the capabilities of state-of-the-art, Ni-base superalloys (e.g. >1000°C–1150 °C). These materials are often used in hotter environments, but rely on sophisticated thermal barrier coating systems to keep the use temperature below this threshold. Promising alternatives for the extension of high temperature capability are available in the refractory domain. Conventional refractory alloys exhibit much higher melting temperatures (e.g. 1800 °C+) and display great promise in retaining suitable mechanical strength at temperatures beyond the melting point of Ni-base superalloys (e.g. 1400 °C+) [1,2].

While many conventional refractory alloys exhibit superior mechanical strength at high temperatures, they are severely deficient in oxidation resistance. It is well known that many inherent refractory oxides are generally deleterious (e.g., Nb_2O_5, Ta_2O_5) and in some cases, highly volatile (e.g., MoO_3, WO_3) [3,4]. This lack of oxidation resistance has been the long-standing inhibitor of the widespread implementation of refractory alloys for high temperature, structural applications. However, the development of new, concentrated and/or high-entropy alloying approaches have enabled material enhancements in both high temperature specific strength and oxidation resistance, amongst other properties [1,5–8]. In particular, concentrated refractory alloys tend to form more complex oxide species (e.g., $CrTaO_4$, $Hf_6Ta_2O_{17}$) generally resulting from the reaction of deleterious oxides with other simple oxides [1,6]. These complex oxides have been reported to induce sluggish oxidation kinetics that may provide reasonable inherent oxidation resistance for short-use and/or expendable applications [6]. However, the influence of compositional complexity and microstructure on the oxidation mechanisms in these alloys are not well understood. Another complicating factor is that balancing mechanical strength, ductility and oxidation resistance is extremely difficult in the refractory domain.

Alloys containing Cr, Nb, Ta and/or Ti show great promise for use in high temperature, structural applications. Nb is highly attractive due to its high melting point (~2477 °C) and low density (8.57 g/cm^3). While Ta is substantially more dense (16.65 g/cm^3), it exhibits a higher melting temperature (~3017 °C) and offers the potential for extended high temperature strength if added in optimal quantities. Both Cr and Ti offer low density and are known to form favorable complex oxides with Nb and Ta [1,6]. In binary space, a substantial body of work has investigated the microstructures and environmental performance of Cr

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(X) + Cr_2X alloys where X is Nb or Ta [9–15]. These alloys form Cr-rich, Cr(Nb) or Cr(Ta) solid solutions in combination with Cr_2Nb or Cr_2Ta Laves phase. The targeted use temperature for these alloys is ~1100 °C. In most cases, the environmental products contain mixtures of simple oxides (e.g., Cr_2O_3, Nb_2O_5) in combination with complex oxides and/or nitrides (e.g., $CrNbO_4$, $TaCrO_4$, Cr_3Nb_3N). Others have described a beneficial influence of Si additions to these alloys [9,14], while Ni, Co, Fe, Al, and Re are reported to offer limited oxidation improvements [13]. Few studies have explored the Nb and/or Ta rich side of these binary alloy systems, which could enable higher temperature use [16–18]. In addition, concentrated alloys in both binary and higher order composition space may offer superior combinations of both strength and environmental resistance.

This article reports the systematic exploration of the oxidation behavior of equiatomic CrNb, CrNbTi, and CrNbTaTi alloys. In particular, the role of compositional complexity on the inherent oxidation mechanisms is addressed.

2. Experimental procedures

Three equiatomic alloys CrNb, CrNbTi, and CrNbTaTi were vacuum arc melted from high purity constituents (99.95%) on a water-cooled copper hearth. A high purity Ti getter was implemented to minimize the extent of residual contamination from oxygen and nitrogen. All alloy buttons were flipped and re-melted several times to promote chemical uniformity throughout the melts. In addition, each button was individually wrapped in Ta foil and hot isostatically pressed (HIP) at 1400 °C for 3 h at 207 MPa under a high-purity argon atmosphere. The heating and cooling rates for HIP were approximately 15 °C/min. The average alloy compositions were consistent with the targeted nominal compositions, as shown in Table 1. Metal concentration was determined using direct current plasma atomic emission spectrometry, carbon was measured using combustion infrared absorption, and oxygen/nitrogen were determined with inert gas fusion. The produced alloys exhibited low concentrations of C, N, and O, Table 1.

Rectangular specimens for microstructural analysis and oxidation testing were sectioned from buttons after HIP using electric discharge machining (EDM). The specimen size was 4.6 mm × 4.6 mm x 7.5 mm. Specimens for microstructural analysis (pre and post oxidation) were hot mounted and prepared using conventional metallographic techniques. Final prep was conducted using a vibratory polisher and 0.05 μm colloidal silica solution. Microstructural analysis was conducted using a combination of X-ray diffraction (XRD), scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS), focused ion beam (SEM-FIB), and transmission electron microscopy (TEM). XRD was conducted on the free surfaces of pre and post oxidation conditions using a tabletop, Bruker D2 Phaser with Cu-Kα1 radiation and 2θ ranging from 20 to 80°. Backscattered electron (BSE) images and "standardless" EDS maps were captured using a TFS Apreo C (HR-SEM) equipped with an Oxford X-max EDS detector. Specimens for TEM analysis were prepared with an FEI Nova dual beam (SEM-FIB) using a variation of the in-situ lift out technique for TEM lamella preparation. High angle annular dark field (STEM-HAADF) images, selected area diffraction patterns (SADPs) and "standardless" EDS maps were captured using an FEI Talos 200X (S)TEM equipped with a super-X, quad EDS detector. Single crystal electron diffraction patterns were analyzed and compared to those simulated using Crystal Maker and Single Crystal software suites.

The reported average EDS values should be considered "standardless" and displayed a standard deviation of no more than 0.5 at.%. For TEM-EDS, a minimum of 3 comparable regions were used to report the average composition for specific phases. In some cases, 5 or more regions were available and used to determine the average value. This was depending on the particular microstructure present in a given TEM foil. For SADPs, a minimum of 3 single crystal patterns were captured and analyzed for each phase. In some cases, 5 or more patterns were captured and used for analysis.

Rectangular specimens for discontinuous oxidation testing were ground to 600 grit, ultrasonically cleaned in acetone followed by isopropanol and dried under pressurized Nitrogen stream. Prior to high-temperature exposure in static lab air, the mass of each specimen was measured using a high sensitivity balance (10^{-6} g) and its dimensions were measured using a caliper (10^{-3} mm). All specimens were placed in yttria stabilized zirconia (YSZ) crucibles and inserted into pre-heated furnaces for 1, 8 and 24 h at 1200 °C in air. Specimen mass was measured post oxidation exposure in order to report specific mass change with respect to initial surface area (mg/cm^2).

2.1. Theoretical methods

Modeling of the oxide products was done using ab initio density functional theory (DFT) via the projector plane wave method implemented in the Vienna Ab Initio Simulation Package (VASP) [19,20] using the Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) of the exchange correlation functional [21]. Several of the complex ternary rutile oxides found experimentally do not have a known cation configuration or lattice parameters. In these cases, the oxides were first modeled as special quasirandom structures (SQS) using the "mcsqs" program implemented in the Alloy-Theoretic Automated Toolkit (ATAT) [22]. $NbTiO_4$, $NbCrO_4$ and $CrTaO_4$ were modeled using known lattice parameters retrieved from the Inorganic Crystal Structure Database [23], while $TaTiO_4$ and $CrTiO_4$ were modeled using the rutile TiO_2 lattice parameters. The best fit structure using a 24-atom unit cell was chosen for each of these oxides as the starting point for DFT simulations.

For all oxides simulated, the plane wave energy cutoff was set to 520 eV. All atoms, bond angles and bond lengths were allowed to relax until the total energy calculations converged to <1 meV/atom, at which point the total energy was calculated with no atomic movement until the energy converged to less than 10^{-4} eV for the entire simulated cell. In the case of oxides containing Cr, the GGA + U method was used [24] to account for the stronger Coulombic interaction of Cr's *d* electrons with the nucleus; the additional energy added using this method was

Table 1
Measured compositions of CrNb, CrNbTi, and CrNbTaTi alloys (at. %). For "Alloy" reported values, carbon was determined by combustion infrared detection; oxygen and nitrogen were determined by inert gas fusion; and all others were determined by DC-plasma emission spectroscopy. All other phase compositions were determined by SEM-EDS alone with an associated error of ≤0.5 at.%.

Alloy	Phase	Cr	Nb	Ti	Ta	C	N	O
CrNb	Alloy	49.9	49.5	–	–	0.012	0.380	0.302
	BCC	19.9	80.1	–	–	–	–	–
	LAVES	62.4	37.6	–	–	–	–	–
CrNbTi	Alloy	32.4	33.6	33.9	–	0.027	0.023	0.137
	BCC	23.4	37.2	39.4	–	–	–	–
	LAVES	59.8	25.3	14.9	–	–	–	–
CrNbTaTi	Alloy	24.5	25.7	25.6	23.9	0.046	0.033	0.161
	BCC	13.4	28.9	31.5	26.2	–	–	–
	LAVES	58.4	9.1	11.5	21.0	–	–	–

corrected after the fact to enable direct comparison with compounds simulated using only GGA, per the guidelines outlined by Materials Project [25,26]. The valence electrons included in the calculations were Nb *4p*, *5s* and *4d*; Ta *5p*, *6s* and *5d*; Cr *3p*, *3d* and *4s*; Ti *3d* and *4s*; O *2s* and *2p*. *K*-point sampling was done using Gamma-centered *k*-point grids with a minimum *k*-point density of 650 *k*-points per reciprocal atom (pra). The cation configuration of the ternary oxide $(Nb,Ta)_{10}O_{25}$ was enumerated using the Clusters Approach to Statistical Mechanics (CASM) software package [27–30], implemented in Python, in order to determine the lowest energy configuration of cations that match the Nb: Ta ratio observed experimentally.

In order to predict the oxide layer structure for each alloy, a model that uses a database of ab initio calculations to determine the composition of the oxide scale as a function of the oxygen chemical potential was employed. This model minimizes the thermodynamic grand potential of oxygen for the system at a given chemical potential of oxygen. Assuming that the system is closed with respect to the metal atoms and open with respect to oxygen, the grand potential of oxygen $\overline{\Phi}_G$ for each compound can be expressed as

$$\overline{\Phi}_G(\mu_O, N_M, P, T) = \frac{G - \mu_O N_O}{N_M} \tag{1}$$

where G is the Gibbs free energy of formation, μ_O is the chemical potential of oxygen, N_M is the number of metal atoms per formula unit, N_O is the number of oxygen atoms per formula unit, P is the pressure, and T is the temperature. In order to find the equilibrium phase compositions, the oxygen grand potential must be minimized. More generally,

$$Min[\overline{\Phi}_G] \text{ subject to : } C_p^T x = C_A, \tag{2}$$

where C_p, x, C_A are the phase composition, phase fraction, and alloy composition matrices, respectively. $C_p{}^T$ has dimensions of $N_{species}$ x N_{phases}, and C_A has dimensions $N_{species}$ x 1. In matrix format:

$$Min[\overline{\Phi}_G] \; subject \; to \; : \; \begin{bmatrix} C_{p11} & \cdots & C_{p1m} \\ \vdots & \ddots & \vdots \\ C_{pn1} & \cdots & C_{pnm} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} C_1 \\ \vdots \\ C_n \end{bmatrix}, \tag{3}$$

where n represents the number of metal species and m represents the number of phases. Thus, the phase fractions x_1, …x_{m-1}, x_m for a given μ_O are the solution to a linear system. Additionally, this methodology can incorporate multiple gaseous species; in this case, nitrogen is also included in the grand potential calculation, with a separate range of chemical potentials specified. The Gibbs free energy of formation for each phase is retrieved from the Materials Project database [31]. Phases that have not been experimentally verified via the Inorganic Crystal Structure Database [23] and phases that are known to be unstable at the given oxidation temperature are removed from the minimization calculation. The code and accompanying Jupyter workbooks to use this methodology are hosted in a GitHub repository [32].

3. Results

3.1. HIP microstructure

The microstructures of CrNb, CrNbTi and CrNbTaTi are reported in great detail elsewhere [33], but are highlighted here to provide context for the oxidation behaviors. X-ray diffraction analysis shows, Fig. 1, that all three studied alloys contain two phases, one of which has a body centered cubic structure (disordered BCC, A2) and another that has an ordered face centered cubic structure (Laves, C15). The lattice parameter of the BCC phase is 326.8 pm for CrNb, 322.8 pm for CrNbTi and 325.0 pm for CrNbTaTi. The lattice parameter of the Laves phase appears to be very similar in all cases and equals 698.3 p.m.

SEM and EDS analysis confirms the X-ray diffraction results by revealing the presence of two distinct phases in these alloys, Fig. 2 and Table 1. On the backscatter electron (BSE) images the Laves phase, which is rich in Cr, appears darker in atomic Z contrast than the BCC phase, which is enriched with the heavier refractory elements of Nb and Ta. The SEM/EDS measured compositions of these phases in the studied alloys are given in Table 1.

The microstructure of the CrNb alloy consists of coarse, near-globular BCC particles and a eutectic mixture of Laves and BCC phases Fig. 2(a) and (d). The eutectic BCC phase is present in the form of discontinuous lamellae (or rods) inside the continuous Laves phase matrix. The microstructures of CrNbTi, Fig. 2(b) and (e), and CrNbTaTi, Fig. 2(c) and (f), are morphologically similar to CrNb. However, the addition of Ti (and subsequent Ta) inverts the microstructure, whereby each consists of a BCC matrix and near-spherical precipitates of the Laves phase, which are present along grain boundaries and inside grains. The BCC phase is depleted in Cr and rich with other alloying elements, while the Laves phase is heavily rich in Cr and can be described as Cr_2Nb or $Cr_2(Nb,Ta)$ Table 1.

3.2. Oxidation behaviors

Linear and logarithmic representations of the oxidation kinetic data for each alloy are shown in Fig. 3 (a) and (b), respectively. In particular, CrNb and CrNbTi displayed similar kinetics out to 24 h at 1200 °C, resulting in total specific mass gains of 14.6 mg/cm^2 and 19.2 mg/cm^2, respectively. The kinetics for CrNbTaTi were higher and resulted in a total specific mass gain of 61.9 mg/cm^2 after 24 h at 1200 °C. It should be noted that the total mass gain considered here is the net behavior of oxidation, volatilization, and nitridation, as described later.

To better understand the relative reaction rates, the kinetic data was

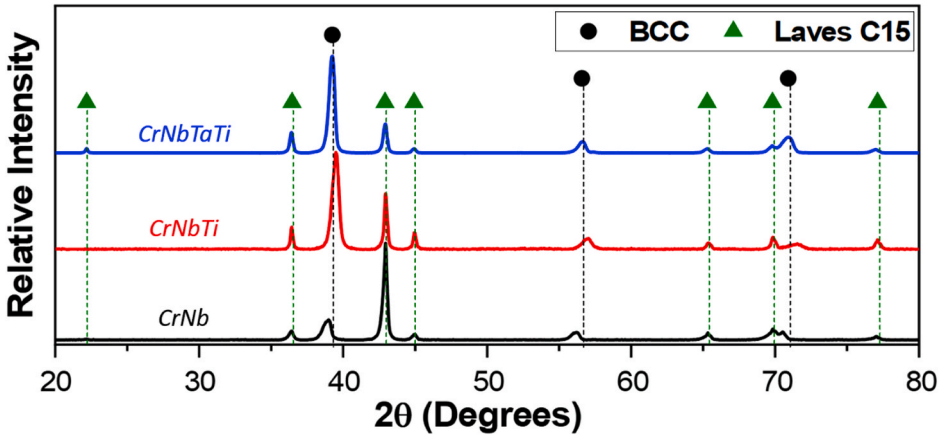


Fig. 1. XRD spectra captured from the HIPed CrNb, CrNbTi, and CrNbTaTi alloys.

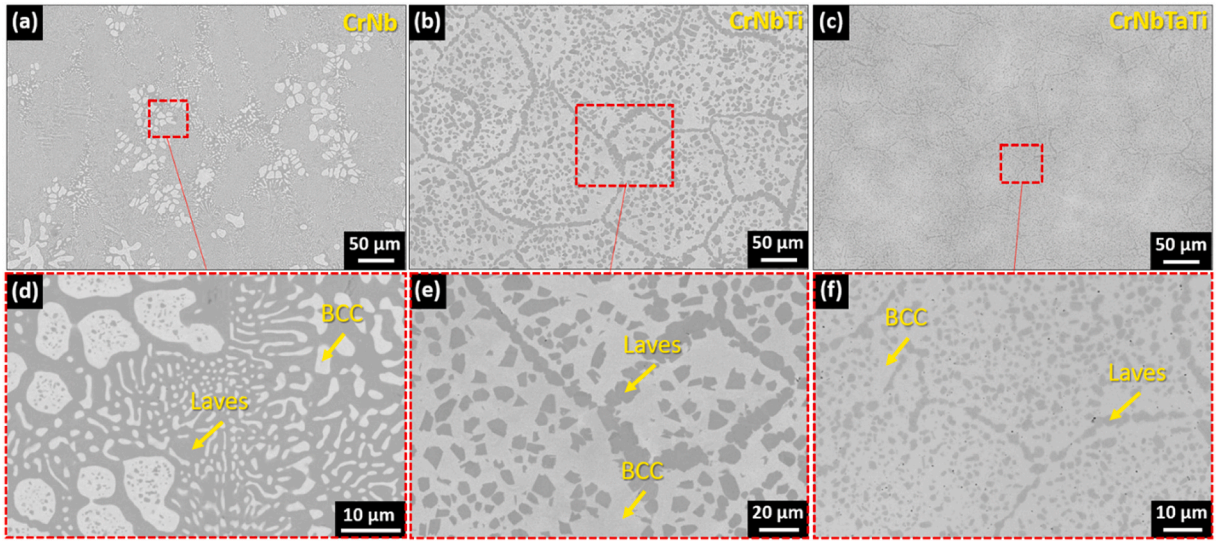


Fig. 2. BSE images of the starting, HIP microstructures for CrNb, CrNbTi, and CrNbTaTi alloys (a)–(b), respectively; and corresponding higher magnification images with phase labels, (d)–(f). To provide the most representative views, all images are not collected at the same magnification.

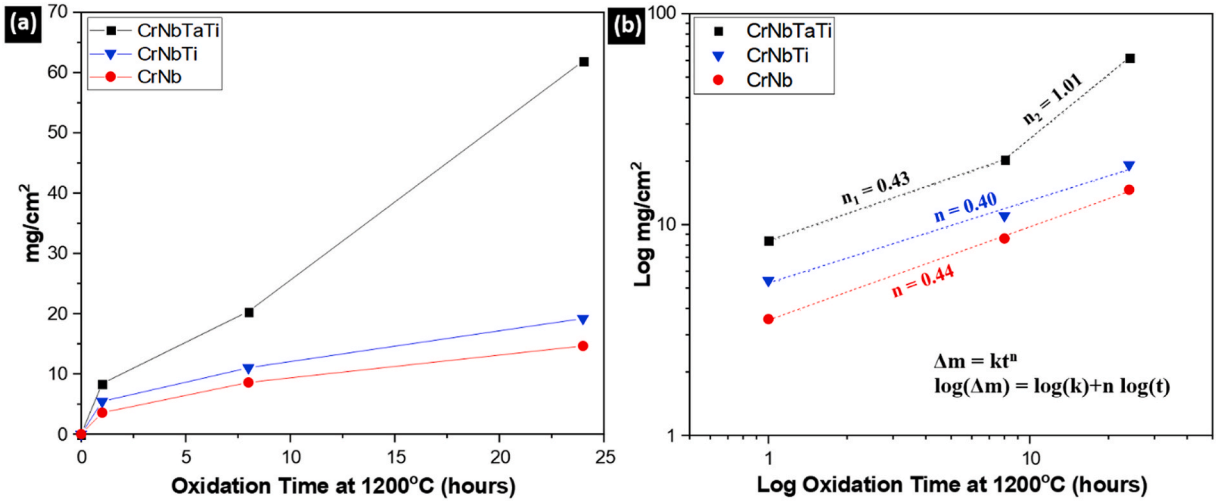


Fig. 3. Specific mass change for CrNb, CrNbTi, and CrNbTaTi during discontinuous oxidation at 1200 °C in air represented in linear form (a) and Log form (b). Dashed lines in (b) represent best-fit according to kinetic parameters shown and listed in Table 2.

analyzed using a variation of the general growth rate law [34]:

$$\Delta m = k\, t^{n} \quad (4)$$

where Δm is specific mass change normalized to initial surface area (e.g. mg/cm^2), k is the rate constant, t is oxidation time and n is the time exponent. It is generally accepted that oxidation kinetics where $n = 1$ are considered "linear" cases that are rate limited by reactions occurring primarily at the oxide-metal interface. Alternatively, oxidation kinetics where $n = 0.5$ are considered "parabolic", where the rate is diffusion controlled either by outward diffusion of cations, inward diffusion of anions and/or a combination of the two types. It is often convenient to represent the oxide growth rate law in log-log form, as shown in Fig. 3 (b). This changes the general form to:

$$\log(\Delta m) = \log(k) + n\log(t) \quad (5)$$

where the y-intercept relates to the rate constant (k) and the slope of the curve is the time exponent (n). It is clear that CrNb and CrNbTi each exhibit one distinct regime of environmental reaction with similar, near-parabolic rate constants (Fig. 3(b); Table 2) of (n = 0.44; k = 3.53 mg cm^{-2} h$^{-0.44}$) and (n = 0.40; k = 5.23 mg cm^{-2} h$^{-0.40}$), respectively. The CrNbTaTi alloy also displays a similar, near-parabolic regime of reaction up to 8 h (n = 0.43; k = 8.40 mg cm^{-2} h$^{-0.43}$) that ultimately transitions to linear behavior (n = 1.01; k = 2.48 mg cm^{-2} h$^{-1.01}$) out to 24 h at 1200 °C. This behavior indicates a shift in mechanism and will be discussed later relative to the observed oxide and nitride morphologies.

Table 2
1200 °C Reaction rate constants (k) and time exponents (n) for CrNb, CrNbTi and CrNbTaTi.

Alloy	Temperature	k (mg cm^{-2} h^{-n})	n	Duration
CrNb	1200 °C	3.53	0.44	0–24hrs
CrNbTi	1200 °C	5.23	0.40	0–24hrs
CrNbTaTi	1200 °C	8.40	0.43	0–8hrs
CrNbTaTi	1200 °C	2.48	1.01	8–24hrs

3.3. Post oxidation microstructures

Fig. 4 shows representative BSE cross-section images of the CrNb,

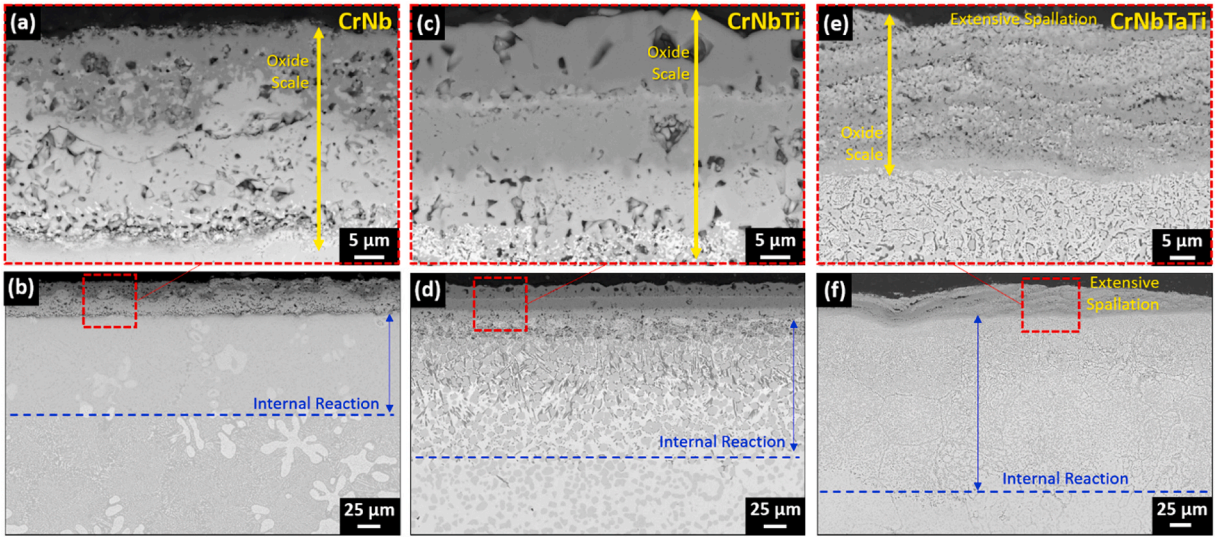


Fig. 4. BSE cross-section images of the CrNb (a)–(b), CrNbTi (c)–(d), and CrNbTaTi (e)–(f) alloys after oxidation at 1200 °C for 8 h in air.

CrNbTi, and CrNbTaTi alloys after 8 h of oxidation exposure at 1200 °C in air. All of the alloys formed oxide scales that were multi-layered and highly complex, Fig. 4 (a), (c), and (e). It should be noted that extensive oxide spallation was observed for CrNbTaTi, thus the oxide scale shown in Fig. 4 (e)–(f) is only a portion of the actual oxide that formed. However, this oxide was contained and accounted for in all mass gain measurements. Each alloy displayed internal reactions zones (IRZ) consisting of internal oxides and extensive nitridation that tended to increase in thickness with respect to alloy elemental complexity, Fig. 4 (b), (d), and (f), respectively. In particular, the IRZ for CrNb was ~106 μm thick and appeared to closely resemble the starting microstructure, Fig. 4(b). However, the overall contrast in these regions was lower than the matrix, indicating pervasive reaction with oxygen and/or nitrogen. For CrNbTi, the IRZ was ~147 μm thick and contained a fine-scale reaction product that also mimicked the starting microstructure, Fig. 4(d). These regions displayed an additional low atomic number Z (low-Z) contrast lenticular phase that tended to form in the prior BCC matrix, in between Laves phase particles. The IRZ for CrNbTaTi was the most extensive with an average thickness of ~181 μm and also consisted of a fine-scale reaction product that resembled the starting microstructure, Fig. 4(f).

Figs. 5–6 shows representative, cross-section SEM-EDS maps and corresponding SEM-EDS line profiles for the CrNb, CrNbTi, and CrNbTaTi after oxidation at 1200 °C for 8 h in air. The alloying elements are grouped together and displayed in composite SEM-EDS maps in Fig. 5(a)–(c), while elemental N and O maps are shown exclusively in Fig. 5(d)–(i). The CrNb alloy exhibited a Cr-rich (green) outer oxide layer on top of an underlying Nb-rich (red) oxide layer, Figs. 5(a) and Fig. 6 (b). These regions exhibited a composition consistent with Cr_2O_3, Fig. 6 (a)–(b). Nb and N enrichment was observed at the interface between the oxide layers and IRZ, as revealed in Fig. 5(a) and (d). Higher N content and some local O-enrichment was also evident in the IRZ, Fig. 5 (d) and (e). Interestingly, the CrNb alloy exhibited little oxygen content at ~40 μm into the sample, yet nitrogen content remained fairly high (~20 at.%) due to extensive nitridation, Fig. 6 (b).

The CrNbTi oxide scale was more complex and consisted of an outer Ti-rich (blue) oxide layer, an underlying Cr-rich (green) oxide layer and a third layer rich in both Nb and Ti, Fig. 5(b). The outer Ti-rich and Cr-rich regions exhibited chemistries consistent with TiO_2 and Cr_2O_3, as shown in Fig. 6 (d). This was followed by the IRZ, comprised of Cr-rich particles surrounded by a Nb-rich matrix that contained Ti-rich lenticular precipitates. Interestingly, *N*-enrichment was observed in both the outer Ti-rich oxide layer, as well as in the Nb-rich regions of the IRZ, Fig. 5 (f). This may indicate that the initial microstructure forms nitrides, which subsequently oxidize with prolonged exposure [11,12,18]. In addition to the outer layers, O-enrichment was also observed in the

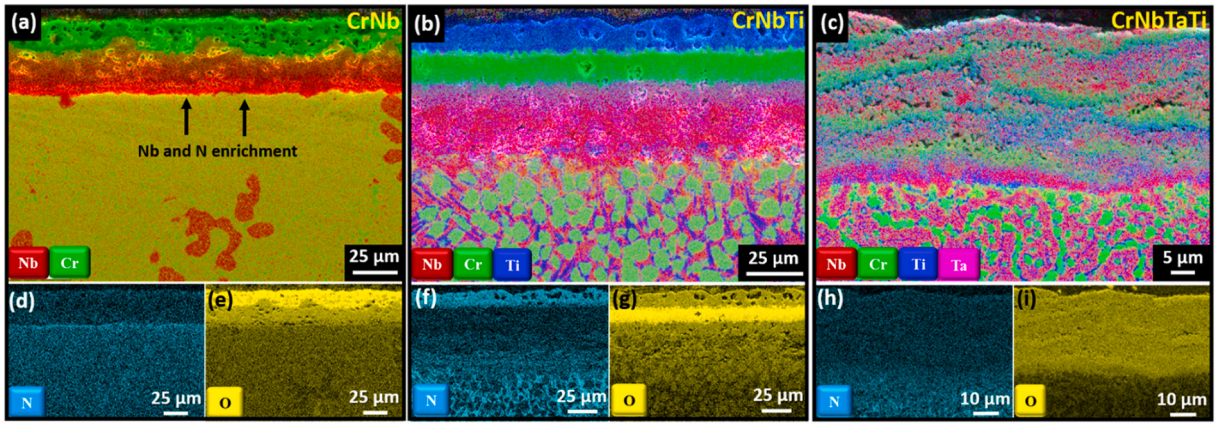


Fig. 5. SEM-EDS cross-section composite maps for CrNb (a), CrNbTi (b) and CrNbTaTi (c), with underlying SEM image, after oxidation at 1200 °C for 8 h in air; and corresponding nitrogen and oxygen elemental EDS maps from the same regions (d)–(e), (f)–(g), and (h)–(i), respectively. To provide the most representative views, all images are not collected at the same magnification.

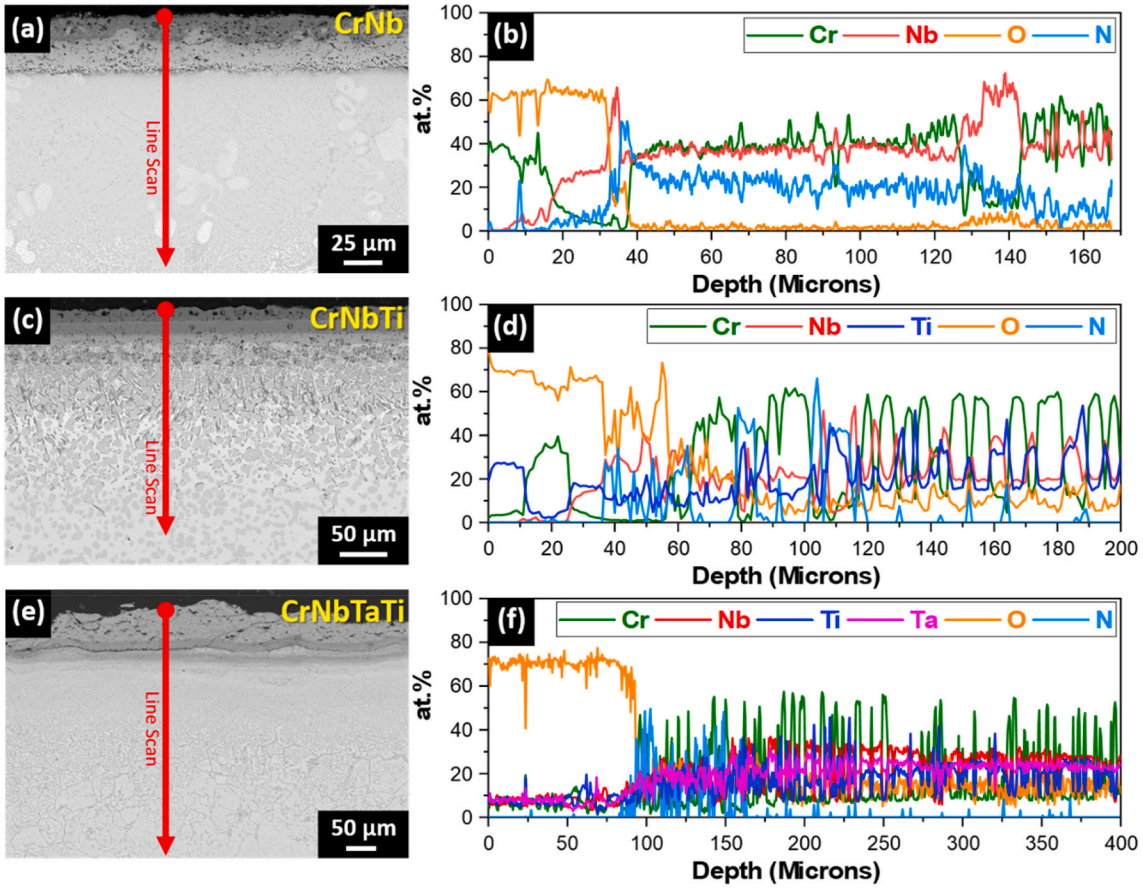


Fig. 6. Backscattered electron images and corresponding SEM-EDS line profiles for CrNb (a)–(b), CrNbTi (c)–(d) and CrNbTaTi (e)–(f), after oxidation at 1200 °C for 8 h in air. The representative locations for the line profiles are shown by the red arrows in (a), (c), and (e). To provide the most representative views, all images are not collected at the same magnification.

IRZ and displayed some preference for the Cr-rich particle phase, Fig. 5 (g).

CrNbTaTi exhibited the most complex oxide morphology, consisting of a discontinuous mixture of several, repeating oxide layers, Fig. 5(c). These convoluted layers displayed enrichment with nearly all alloying elements, Fig. 6 (f). This was followed by a transition into the fine-scale IRZ product that was similar to the coarser structure observed for CrNbTi, Fig. 5(b)–(c). *N*-enrichment was observed primarily in the IRZ for CrNbTaTi, while O-enrichment was found mostly in the outer oxide layers and to some extent, localized to Laves particles in the IRZ, Fig. 5

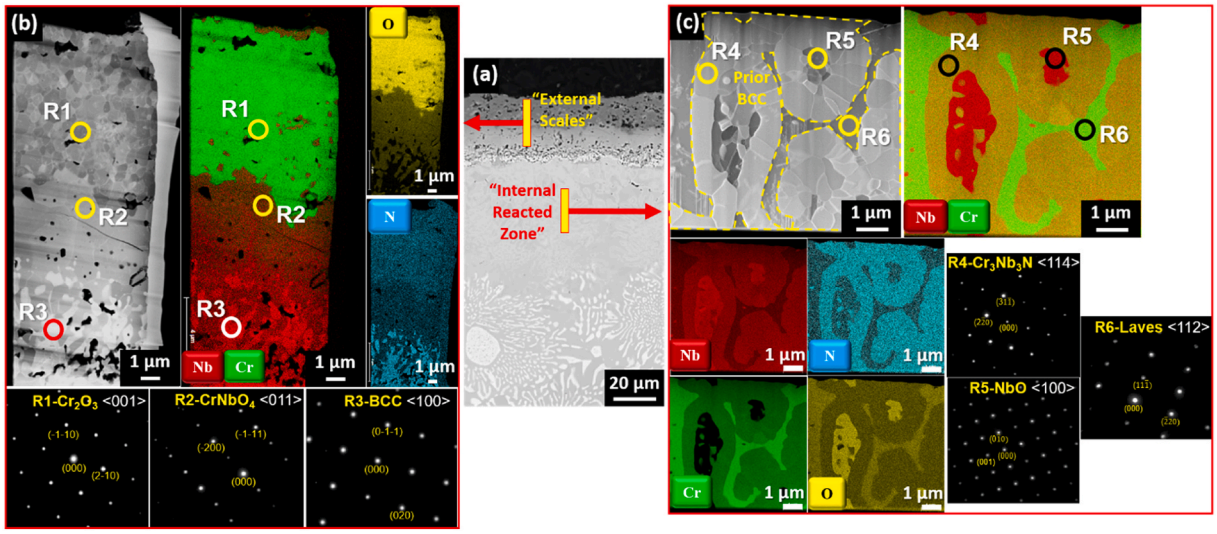


Fig. 7. BSE SEM image of the TEM lift-out sites for CrNb after oxidation at 1200 °C/8hrs in air (a); corresponding STEM-HAADF images, composite STEM-EDS maps, elemental TEM-EDS maps, and representative selected area diffraction patterns from the phases present for the "external scales" (b) and "internal reacted zone" (c), respectively.

(h)–(i).

TEM analysis was conducted on the oxide structures that formed on the CrNb alloy after 8 h of oxidation at 1200 °C in air. Fig. 7 shows a BSE image revealing the TEM lift-out sites (a), corresponding STEM-HAADF images, STEM-EDS maps, and SADPs from the external scales (b) and IRZ (c), respectively. Similar to the SEM observations, the outer scale consisted of an equiaxed, fine-grain, Cr-rich phase (R1) on top of a Nb-rich layer (R2) containing Nb and *N*-rich precipitates (R3) further into the alloy, Fig. 7(b). The Cr-rich phase (R1) exhibited a structure consistent with Cr_2O_3 (*R-3c*; SG 167), as evident by the SADP in Fig. 7 (b), and STEM-EDS revealed enrichment in both O and Cr, Table 3. R2 was enriched in Nb, Cr, and O, with ~1.5 times the concentration of Nb to Cr, Table 3. Structural analysis revealed that R2 displayed a structure consistent with $CrNbO_4$ (*P42/m n m*; SG 136), as shown by the SADP in Fig. 7(b). This structure is analogous to rutile (i.e., TiO_2), but assumes random, shared occupancy on the cation sublattice between Nb and Cr. The smaller precipitates (R3) were found to be extensively rich in Nb, with around 5 at.% N and trace amounts of Cr and O, Table 3. These precipitates displayed a body centered cubic (BCC) structure (*Im-3m*; SG 229) and are likely remnant BCC phase from the starting microstructure, Fig. 7(b). It is anticipated that these regions would likely transform to Nb-based nitrides (e.g., Nb_2N, NbN) at longer exposure durations, but additional TEM is required to justify this hypothesis. These structures were observed for the CrNbTi and CrNbTaTi, as reported later.

A similar analysis was conducted on the IRZ of the CrNb alloy after 8 h of oxidation at 1200 °C in air, Fig. 7. This region displayed prominent chemical segregation and a total of 3 distinct regions that resembled the initial BCC particles in the Laves phase matrix, Fig. 2(d). Prior BCC particles were outlined with an equiaxed, Nb + Cr + N-rich phase (R4) that ultimately surrounded a Nb + O-rich center core (R5). Nb + Cr + O-rich channels (R6) between these particles were evident and resembled the initial Laves phase matrix, Fig. 7(c). R4 was found to exhibit high concentrations of both Nb and Cr in near equal proportion, along with ~14 at.% N and a small amount of O, Table 3. This region (R4) displayed a structure consistent with Cr_3Nb_3N (*Fd-3m*; SG 227), Fig. 7(c). This complex nitride has been previously observed in the oxidation/nitridation of Cr(X)+Cr_2(X) alloys where X = Nb or Ta [11]. The center cores (R5) were extensively rich in Nb and O with similar concentrations, Table 3, and exhibited a structure consistent with NbO (*Pm-3m*; SG 221), Fig. 7(c). While NbO is not as commonly observed as the pentoxide (i.e., Nb_2O_5) in the high temperature oxidation of Nb-based alloys, it is still the most thermodynamically stable and is likely to form internally where the oxygen partial pressure is considerably reduced compared to the free surface. The remaining matrix (R6) had a composition close to Cr_2Nb, along with ~7 at.% O and trace amounts of N, Table 3. R6 was found to exhibit a diamond-face-centered cubic structure (*Fd-3m*; SG 227) and is believed to be O-enriched prior C15 Laves phase from the initial microstructure, Fig. 2(d). While oxygen was mostly spent by the formation of the external scales, it is clear that some amount does diffuse deeper into the alloy to create the IRZ in concert with nitrogen.

The 1200 °C primary oxide structures for CrNb were also validated using XRD, as shown in Fig. 8(a). XRD spectra were captured from the free surfaces of the oxidized specimens to emphasize the near surface oxide layers. In agreement with TEM analysis, the 1 h, 8 h, and 24 h conditions all exhibited peaks associated with Cr_2O_3 and $CrNbO_4$. This suggests that the dual layer oxide product likely forms early on and is sustained over longer oxidation exposures. This is also in agreement with the kinetic data that revealed a single regime of environmental reaaction throughout the entire extent of exposure, Fig. 3(b).

Table 3
STEM-EDS (at.%) for the phases observed in the CrNb alloy after 8 h of oxidation at 1200 °C. Region and prototype designations correspond to the nomenclature in Fig. 6. Phase compositions are based on the average of at least 3 comparable regions and displayed an associated error of ≤0.5 at.%.

Region	Region/Prototype	Nb	Cr	O	N
External Oxides	**R1**/Cr_2O_3	0.1	37.2	62.7	–
	R2/$CrNbO_4$	20.2	13.6	65.7	0.6
	R3/BCC	91.2	2.2	1.8	4.9
Internal Reaction Zone	**R4**/Cr_3Nb_3N	38.2	42.0	5.9	13.9
	R5/NbO	44.6	2.2	52.0	1.2
	R6/Laves	30.5	61.7	7.0	0.7

TEM analysis was conducted on the CrNbTi alloy after 8 h of oxidation at 1200 °C. Figs. 9–11 and Table 4 show data captured from the outer oxide layers, the transition zone between the outer oxide/IRZ and near the internal oxidation front. The outer oxide layers consisted of "rutile" structured (*P42/m n m*; SG 136) TiO_2 (R1), a secondary Cr_2O_3 (*R-3c*; SG 167) layer (R2), and an underlying Nb + Ti rich, "rutile" structured $NbTiO_4$ (*P42/m n m*; SG 136) layer, analogous to TiO_2 (R3). R3 also contained Nb + N-rich, Nb_2N (*P63/mmc*; SG 194) precipitates (R4), with an ~2:1 ratio of Nb to N, Fig. 9(b). It should be noted that an additional layer between TiO_2 (R1) and Cr_2O_3 (R2) was also observed, as indicated by the yellow arrow in the STEM-EDS composite map in Fig. 9 (b). While this region was too small for selected area electron diffraction, STEM-EDS line scans of the metal elements revealed significant enrichment in Nb, with secondary enrichment in Ti and tertiary enrichment in Cr, Fig. 9(b).

The 1200 °C surface oxide layers for CrNbTi were evaluated using XRD, as shown in Fig. 8(b). Consistent with TEM observations, the 1 h, 8 h, and 24 h conditions all contained peaks associated with $NbTiO_4$, Cr_2O_3, and TiO_2. In addition, peaks associated with $TiNb_2O_7$ were evident and became less pronounced at longer oxidation exposure times, Fig. 8(b). This phase is known to form in binary Nb–Ti alloys a temperatures above 1000 °C [35]. It is thought that this phase correlates to the Nb + Ti-enriched region that was observed between the outer TiO_2 layer and underlying Cr_2O_3 layer, Fig. 9(b). Under short oxidation durations, it is hypothesized that the individual oxides layers are thin and this Nb + Ti-enriched region likely constitutes a larger majority of the scale. However, as the scale grows with extended exposure, the XRD signatures suggest that the $TiNb_2O_7$ becomes a minor phase, which is consistent with the TEM observations after 8 h of oxidation, Fig. 9(b). Further TEM work exploring shorter oxidation conditions is required to validate this concept.

The region between the outer oxide and IRZ for CrNbTi was quite complex, consisting of pores and several chemically distinct phases, Fig. 10. Ti + O-enriched regions (R1) and Nb + Cr + N-enriched regions (R2) were observed throughout the entire TEM foil, as shown by the composite STEM-EDS maps in Fig. 10(b). In addition, Nb + N-enriched regions (R3) were found to exclusively reside in "prior BCC" regions, while Cr + O-enriched (R4) precipitates were observed only in conglomerations believed to resemble "prior Laves" phase particles, as shown in the STEM-EDS composite maps and Table 4. Prior Laves phase boundaries were estimated and are shown schematically with dashed lines in Fig. 10(b). The Ti + O-enriched regions were "rutile" structured (*P42/m n m*; SG 136) TiO_2 (R1), while the Nb + Cr + N-enriched regions were structurally and chemically consistent with Cr_3Nb_3N (*Fd-3m*; SG 227), Fig. 10(b) and Table 4. This is the same complex nitride that was also observed in the IRZ of CrNb, Fig. 7(c). The Nb + N-enriched regions (R3) that were observed exclusively in the prior BCC regions were Nb_2N (*P63/mmc*; SG 194), Fig. 10(b) and Table 4. This nitride is analogous to the nitride phase observed in the outer oxide scales for CrNbTi, Fig. 9(b). Lastly, the Cr + O-enriched precipitates (R4) that were exclusively observed in prior Laves phase regions were consistent with Cr_2O_3 (*R-3c*; SG 167), Fig. 10(b) and Table 4.

The internal oxidation front for CrNbTi contained a Nb-rich BCC (*Im-3m*; SG 229) matrix (R1) with highly faceted TiN (*Fm-3m*; SG 225) precipitates (R2), Fig. 11(b) and Table 4. The BCC phase was depleted in Ti and Cr compared to the initial BCC phase, likely due to the selective reaction of Ti and Cr, Table 4. In addition, these regions shared boundaries with large Cr-rich, C15 Laves phase (*Fd-3m*; SG 227) islands (R3). Nitrogen enrichment was predominantly observed in the Ti-rich (R2) regions, while oxygen was concentrated in the Cr-rich (R3)

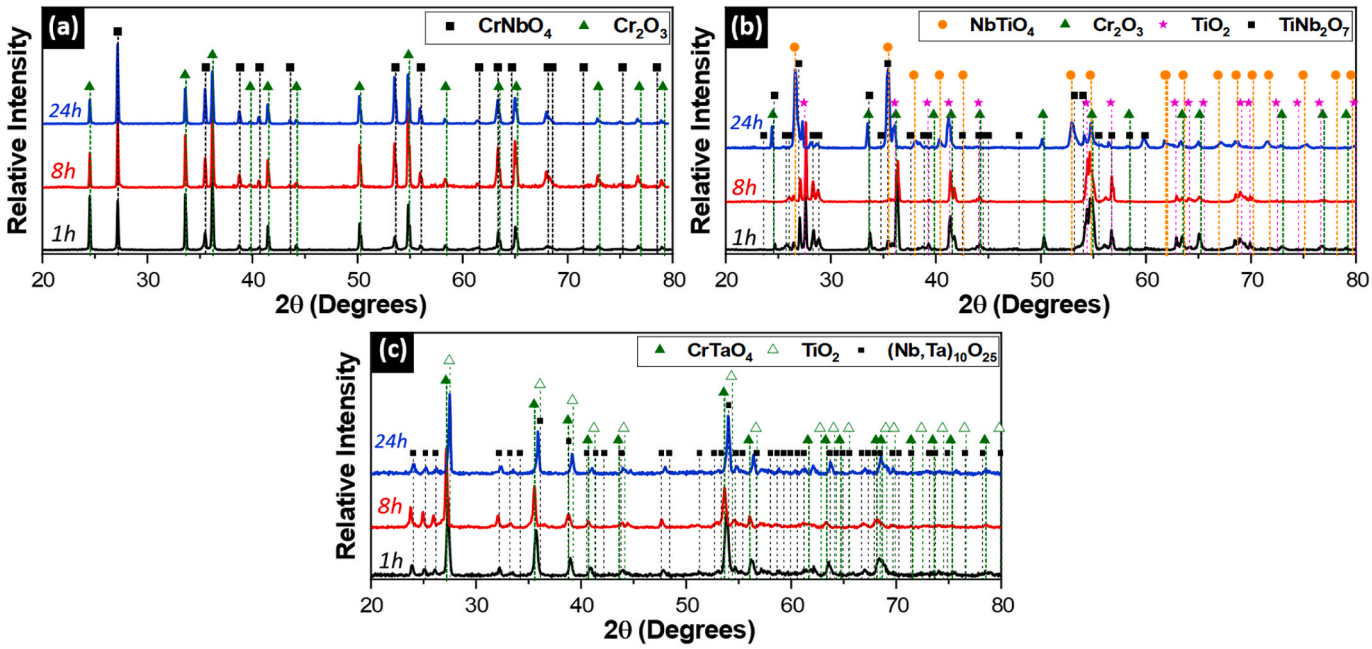


Fig. 8. XRD spectra captured from the CrNb (a), CrNbTi (b), and CrNbTaTi (c) alloys after oxidation at 1200 °C in air.

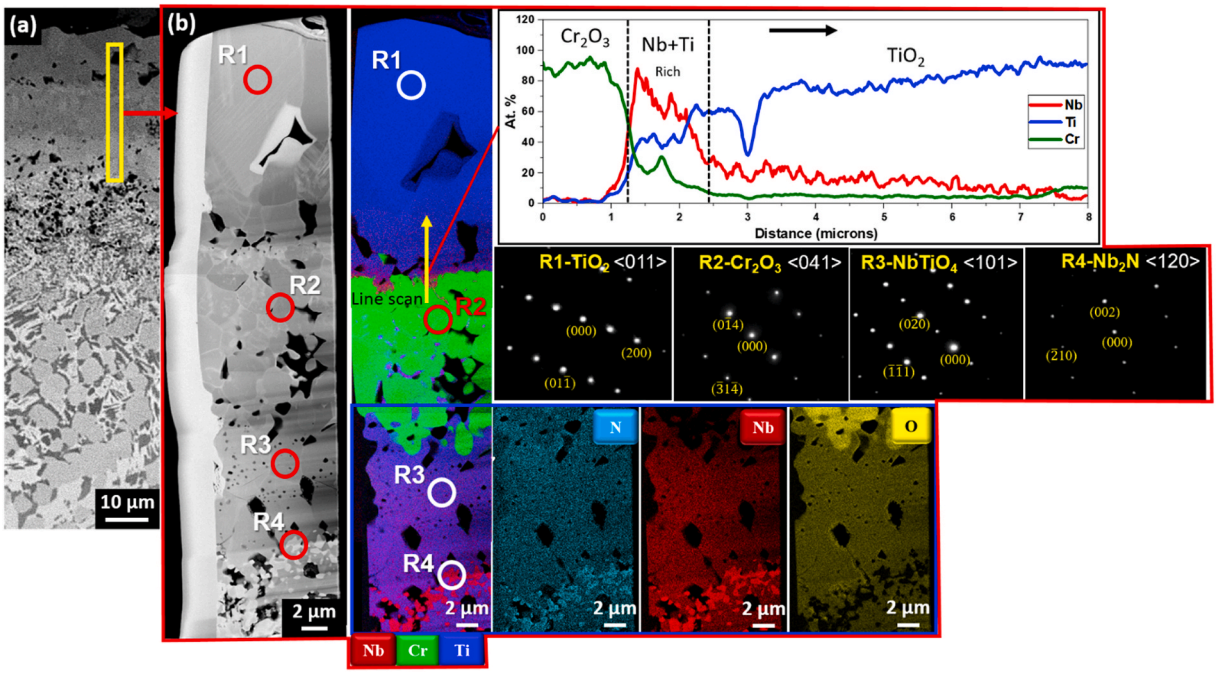


Fig. 9. BSE SEM image of a TEM lift-out site for CrNbTi after oxidation at 1200 °C/8 h in air (a); and corresponding STEM-HAADF image, composite/elemental STEM-EDS maps, representative selected area diffraction patterns from the phases present and STEM-EDS line scan data. The lift-out site was from the outer most oxide.

regions, as shown by the elemental STEM-EDS maps in Fig. 11(b).

Fig. 12 (a)-(b), (c)-(d), and (e)-(f) show cross-section BSE images and composite SEM-EDS maps of the CrNbTaTi alloy after oxidation at 1200 °C for 1 h, 8 h, and 24 h, respectively. After 1hr of oxidation, this alloy formed a distinct, repeating layered structure consisting of selective enrichment from all alloying elements, Fig. 12 (a)–(b). This initial structure also displayed a significant amount of cracks/pores between layers, suggesting an unfavorable stress state. It is likely that temperature induced thermal expansion incompatibility between the alloy and oxides (e.g. Pilling-Bedworth ratio) aids in the formation of cracks [36, 37]. In addition, dissolution of oxygen into the alloy [38] and the formation of internal oxides and nitrides [39] can also aggravate the overall stress state. With prolonged exposure, this layered oxide grew in thickness and appeared to maintain some level of repeating structure, Fig. 12 (e)–(f). However, the extent of distinct layering tended to diminish over longer length scales, as shown in the 24hr composite SEM-EDS map in Fig. 12(f). In addition, the extent of cracking increased and large through thickness voids were observed, Fig. 12(e)–(f).

Fig. 13 shows a composite SEM-EDS map (a), corresponding BSE image (b) and SEM-EDS line scan data for the 1200 °C/1 h repeating

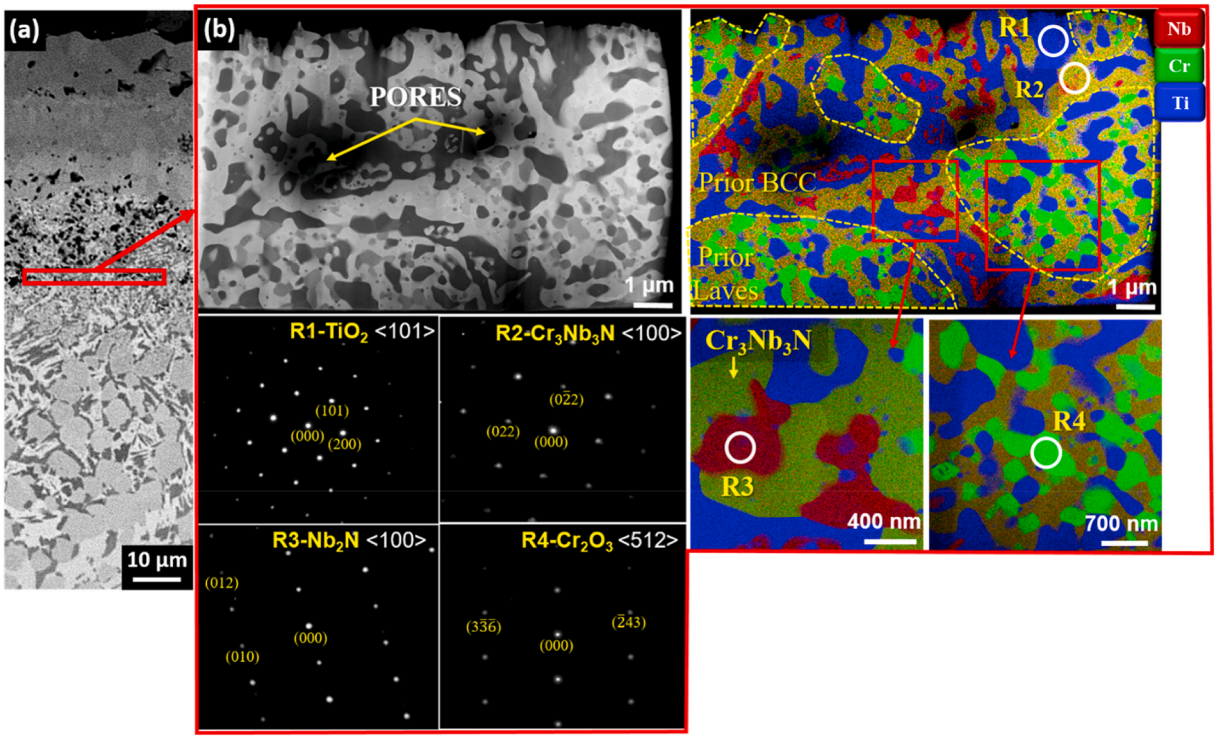


Fig. 10. BSE SEM image of a TEM lift-out site for CrNbTi after oxidation at 1200 °C/8 h in air (a); and corresponding STEM-HAADF image, composite STEM-EDS maps and representative selected area diffraction patterns from the phases present. The lift-out site was from outer oxide/IRZ transition zone.

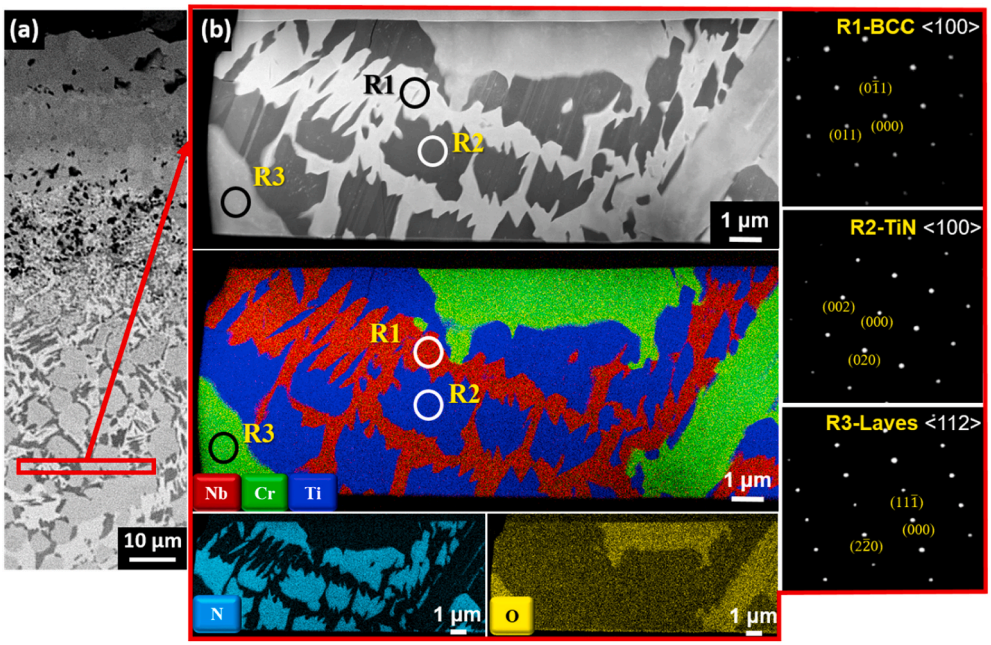


Fig. 11. BSE SEM image of a TEM lift-out site for CrNbTi after oxidation at 1200 °C/8 h in air (a); and corresponding STEM-HAADF image, composite/elemental STEM-EDS maps and representative selected area diffraction patterns from the phases present. The lift-out site was from the internal oxidation front.

oxide structure (free surface is on the left). The SEM-EDS line scan of metal elements revealed a sequence of three repeating layers and captured nearly 2 periods of this pattern, Fig. 13(c). The layers consisted of a Cr-rich layer, a Ti-rich layer and a Nb + Ta-rich layer. However, in all cases, the degree of enrichment was only minor, as all layers contained significant amounts of at least one other alloying element, Fig. 13 (c). Interestingly, the oxygen content was similar in all layers (~66–69 at. %).

TEM analysis was also conducted on the oxide structures that formed on the CrNbTaTi alloy after 8 h of oxidation at 1200 °C in air. Since this alloy exhibited extensive oxide spallation, any oxide products at the outer surface should be considered remnant scale, as opposed to the true free surface of the oxide scale. The outer layers at the oxide/IRZ interface consisted of a variety of chemically distinct layers, as shown by the composite STEM-EDS map in Fig. 14(b). The outermost portion contained a mixture of Cr + Ta + Ti + O-enriched phase (R1), along with

Table 4
STEM-EDS (at.%) for the phases observed in the CrNbTi alloy after 8 h of oxidation at 1200 °C. Region and prototype designations correspond to the nomenclature in Figs. 9–11. Phase compositions are based on the average of at least 3 comparable regions and displayed an associated error of ≤0.5 at.%.

Region	Region/ Prototype	Nb	Cr	Ti	O	N
External Oxides *(Fig. 9)*	**R1**/TiO_2	–	3.6	28.7	67.8	–
	R2/Cr_2O_3	0.1	36.2	0.4	63.4	–
	R3/$NbTiO_4$	16.4	6.1	14.6	63.0	–
	R4/Nb_2N	64.7	1.6	1.0	3.2	29.6
1st Internal Reaction Zone *(Fig. 10)*	**R1**/TiO_2	0.1	1.8	31.2	66.9	–
	R2/Cr_3Nb_3N	38.7	44.2	0.8	1.7	14.6
	R3/Nb_2N	66.3	0.6	1.8	1.3	30.0
	R4/Cr_2O_3	1.6	39.4	0.2	58.8	–
2nd Internal Reaction Zone *(Fig. 11)*	**R1**/BCC	62.3	16.3	14.8	0.8	5.8
	R2/TiN	1.5	0.2	45.0	1.2	52.2
	R3/Laves	16.9	60.6	17.4	5.1	–

Nb + Ta + O-enriched phase (R2), Fig. 14(b) and Table 5. R1 regions were found to be structurally consistent with $(Cr,Ta,Ti)_2O_4$ (*P42/m n m*; SG 136), Fig. 14(b). This oxide is analogous to the TiO_2, $CrNbO_4$ and $NbTiO_4$ oxides observed in the other alloys and assumes that Cr, Ta, and Ti are substituting for each other. The Nb + Ta + O-enriched regions (R2) exhibited a structure consistent with $(Nb,Ta)_{10}O_{25}$ (*I4/m*; No. 87), Fig. 14(b). This particular oxide is known to form at high temperatures in systems where Nb_2O_5 and Ta_2O_5 oxides are prevalent [40]. These regions were followed by another layer (R3) that exhibited a composition similar to R1 followed by Cr + O-enriched precipitates (R5), Table 5. While SADPs were not trivial from these regions, chemical analysis suggests that R3 may exhibit a $(Cr,Ta,Ti)_2O_4$ structure, while the Cr + O-enriched precipitates (R4) are likely Cr_2O_3, Table 5. Below these Cr-rich regions, another layer consisting of Cr + Ta + Ti + O enrichment (R5) was observed, Table 5. Structural analysis revealed that R5 regions exhibited a $(Cr,Ta,Ti)_2O_4$ (*P42/m n m*; SG 136) structure, as shown by the SADP in Fig. 14(b). Below R5, another region exhibiting a composition similar to Cr_2TiO_5 (R6) was observed, Table 5. This region was structurally interrogated in a subsequent TEM foil and will be discussed later.

The 1200 °C surface oxide layers for CrNbTiTa were evaluated using XRD, as shown in Fig. 8(c). Consistent with TEM observations, the 1hr, 8hr, and 24hr conditions all contained peaks associated with (Nb, $Ta)_{10}O_{25}$. In addition, broad peaks associated with "rutile" structured complex oxides were also observed. Based on the TEM analysis, it is clear that the CrNbTaTi alloy formed a variety of these rutile structure oxides with various ratios of Cr, Ta, and Ti, Fig. 14(b) and Table 5. The ratios of these elements in the oxide scale ultimately govern the lattice parameters associated with the respective space group (*P42/m n m*; SG 136). To illustrate the "bounds" of possible lattice parameters within these structures, the XRD plot in Fig. 8(c) shows peaks associated with TiO_2 and $CrTaO_4$. TiO_2 exhibits the minimum lattice parameters (a = 4.593 Å, c = 2.959 Å) due to its small atomic radius (147 pm), while $CrTaO_4$ displays larger lattice parameters (a = 4.644 Å, c = 3.009 Å) due to the atomic radii of Cr (200 pm) and Ta (220 pm). It is hypothesized that the range of rutile, complex oxides observed in the CrNbTaTi alloy likely fall within these constraints.

Fig. 14(c) shows the phases captured from the IRZ of the CrNbTaTi alloy. Similar to R6 from the oxide/IRZ interface, Fig. 14(b), the top of the IRZ contained a Cr + Ti + O-enriched, Cr_2TiO_5 (*C2/c*; SG 15) region (R7) followed by a mixture of Ta + Ti + O-enriched, $TaTiO_4$ (*P4₂/m n m*; SG 136) regions (R8) interwoven with Nb + Ta + N-enriched, Nb_2N (*P6₃/mmc*; SG 194) regions (R9), Fig. 14(c) and Table 5. Based on the composition of the Nb_2N regions it is assumed that Nb and Ta share occupancy in the form $(Nb,Ta)_2N$. Deeper into the IRZ, extensively Cr-rich regions (R10) were observed, Fig. 14(c) and Table 5. Although SADPs were not captured from these regions, a chemical inference would suggest that they are remnant Laves phase from the initial microstructure that has started reacting with oxygen and/or nitrogen, Table 5.

The 1200 °C/8 h microstructure of the CrNbTaTi alloy near the internal oxidation front was also investigated and contained similar chemical segregation compared to the observations made for the internal oxidation front for the CrNbTi alloy, Figs. 11(b) and Fig. 15(b). However, the microstructure in this region for the CrNbTaTi alloy was much finer, due to the size scale of the starting microstructure. In particular, the foil contained a continuous Nb + Ta-rich, BCC (*Im-3m*; SG 229) matrix (R2) with a pervasive distribution of TiN (*Fm-3m*; SG 225) precipitates (R3). Additionally, larger Cr + Ta-rich, Laves C15 (*Fd-3m*; SG 227) particles (R1) were also present and formed similar nitrides along boundaries. This behavior suggests a potential ejection of Ti from the Laves phase to form bounding TiN. Concurrently, the BCC phase exhibits a substantially high fraction of Ti (~30 at. %) and therefore, precipitates out TiN pervasively.

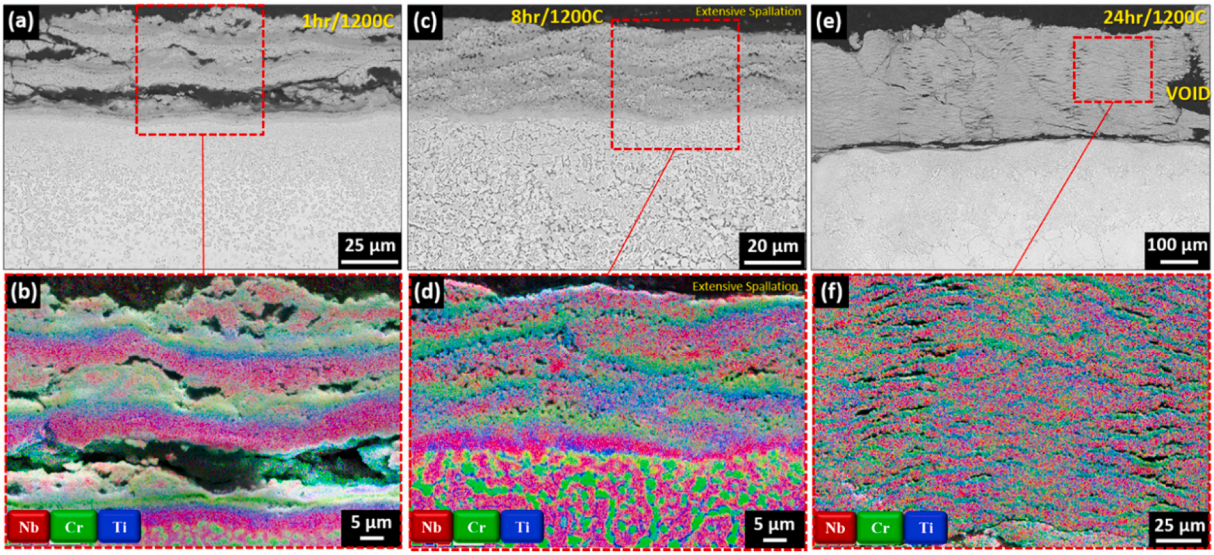
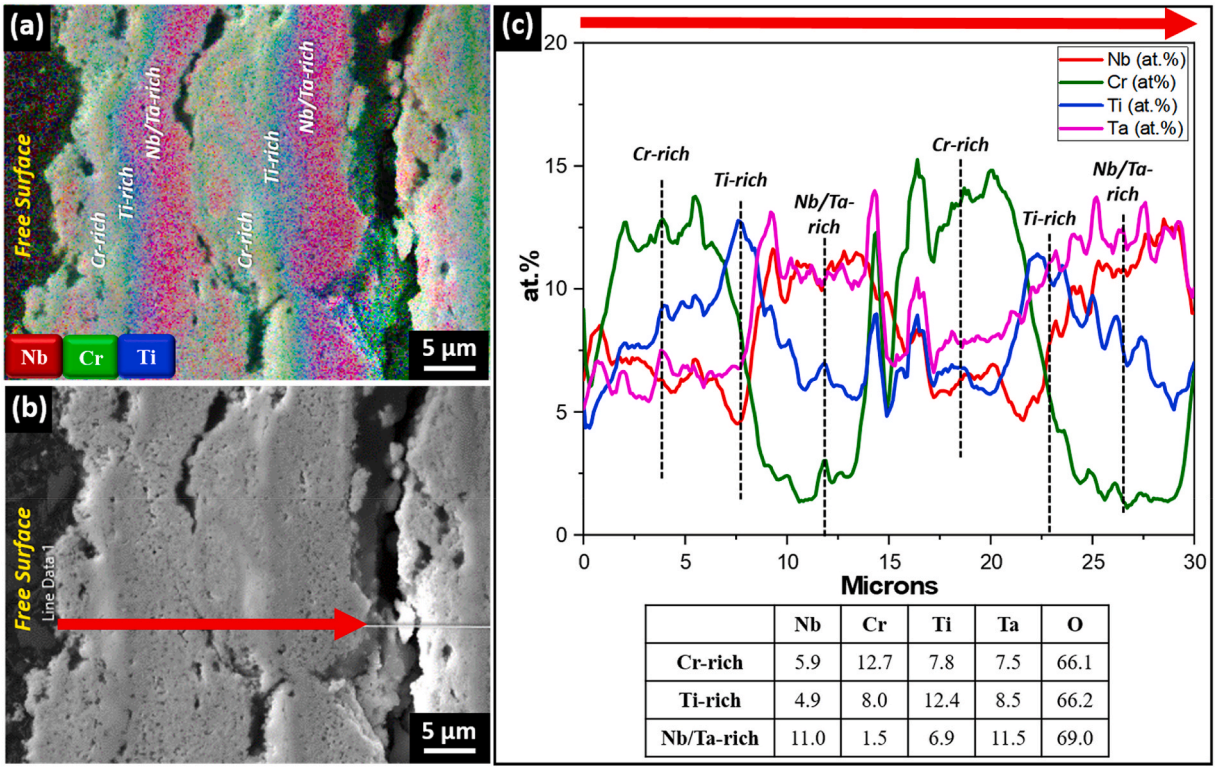


Fig. 12. BSE cross-section images and corresponding, inset composite SEM-EDS maps of the CrNbTaTi alloy after oxidation at 1200 °C for 1 h (a)–(b), 8 h (c)–(d), and 24 h (e)–(f), respectively. To provide the most representative views, all images are not collected at the same magnification. In addition, the EDS maps were captured from the areas inset by red boxes in (a), (c), and (e).



	Nb	Cr	Ti	Ta	O
Cr-rich	5.9	12.7	7.8	7.5	66.1
Ti-rich	4.9	8.0	12.4	8.5	66.2
Nb/Ta-rich	11.0	1.5	6.9	11.5	69.0

Fig. 13. BSE cross-section image and composite SEM-EDS map of the CrNbTaTi alloy after oxidation at 1200 °C for 1 h (a)–(b), along with SEM-EDS line scan of metal cations and corresponding average compositions in each layer (c). The arrows in (b) and (c) represent the location and direction of the line scan data (into the alloy).

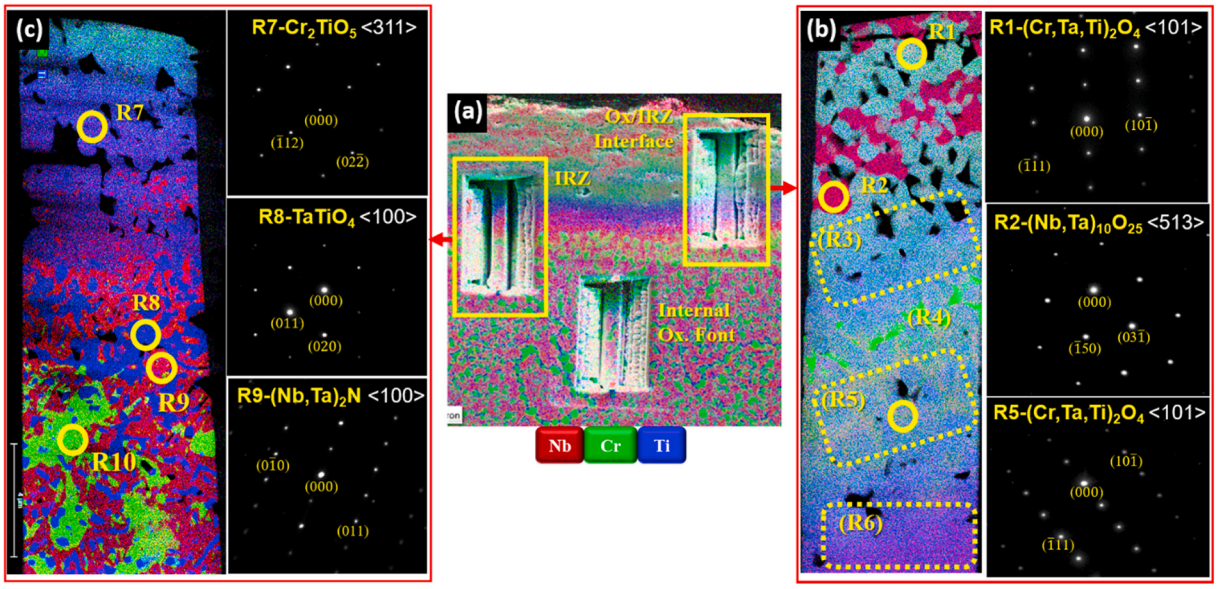


Fig. 14. Composite SEM-EDS map depicting the TEM lift-out sites for CrNbTaTi after oxidation at 1200 °C/8 h in air (a); corresponding STEM-HAADF images, composite STEM-EDS maps, and representative selected area diffraction patterns from the phases present for the oxide/IRZ interface (b) and IRZ (c), respectively.

3.4. Formation energies and thermodynamic stability of oxides

The experimentally observed oxides were simulated using DFT to calculate the formation energies and their minimum energy above the convex hull. These formation energies, referenced against the chemical potentials of the pure elements, are shown in Table 6, and were included in the oxide layer composition predictions detailed in the next section. Additionally, the minimum energy above the convex hull was calculated for all phases over a range of μ_N and μ_O that were determined to best match the experimental characterization results (detailed in Section 3.5). All energy above hull values less than 10^{-6} eV/atom are assumed to be on the convex hull. Most of the mixed ternary rutile oxides sit on the convex hull, or extremely close to it. It can be concluded that these phases are thermodynamically stable, not metastable. The two notable exceptions to this are $NbCrO_4$ and $NbTiO_4$, which are 350 meV/atom and 90 meV/atom above the convex hull, at a minimum, respectively. It

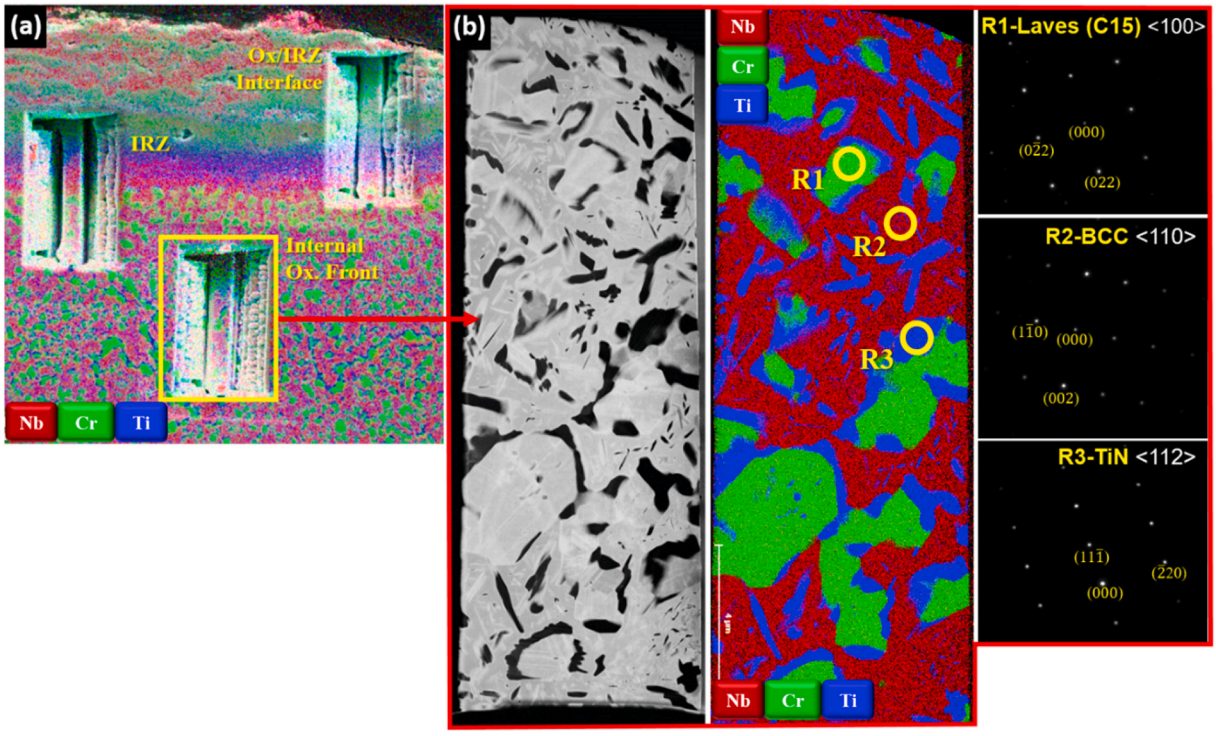


Fig. 15. Composite SEM-EDS map of a TEM lift-out site for CrNbTaTi after oxidation at 1200 °C/8 h in air (a); and corresponding STEM-HAADF image, composite STEM-EDS map and representative selected area diffraction patterns from the phases present. The lift-out site was near the internal oxidation front.

Table 5
STEM-EDS (at.%) for the phases observed in the CrNbTaTi alloy after 8 h of oxidation at 1200 °C. Region and prototype designations correspond to the nomenclature in Figs. 14–15. Phase compositions are based on the average of at least 3 comparable regions and displayed an associated error of ≤0.5 at.%. *(Asterisks represent phase prototypes that are anticipated based on chemical inference alone)*.

Region	Region/Prototype	Nb	Cr	Ti	Ta	O	N
Oxide/IRZ Interface *(Fig. 14(b))*	**R1**/$(Cr,Ta,Ti)_2O_4$	4.5	13.1	6.1	10.6	65.7	–
	R2/$(Nb,Ta)_{10}O_{25}$	16.6	0.9	6.6	11.3	64.6	–
	R3/$(Cr,Ta,Ti)_2O_4$*	3.0	11.7	8.8	13.9	62.7	–
	R4/Cr_2O_3*	1.7	39.1	1.9	2.5	54.8	–
	R5/$(Cr,Ta,Ti)_2O_4$	1.0	13.6	12.1	10.2	63.1	–
	R6/Cr_2TiO_5*	1.7	23.5	12.4	1.5	60.9	–
IRZ *(Fig. 14(c))*	**R7**/Cr_2TiO_5	3.7	23.2	12.6	1.6	58.9	–
	R8/$TaTiO_4$	2.0	1.6	18.7	13.1	64.6	–
	R9/$(Nb,Ta)_2N$	31.4	1.0	3.6	28.4	6.6	29.0
	R10/Laves*	11.0	58.9	3.2	15.3	3.6	8.2
Internal Ox. Front *(Fig. 15)*	**R1**/Laves	8.9	62.5	3.7	22.7	1.5	0.7
	R2/BCC	41.8	4.8	9.1	38.7	1.9	3.9
	R3/TiN	2.1	2.5	42.4	4.1	3.0	45.9

is likely that these structures are either stabilized via high temperature vibration effects or are kinetically favored to form quickly, but are then trapped in a metastable state. At higher temperatures, the mixed ternary rutile oxides are also likely mutually soluble due to similar lattice parameters, which would result in the complex quaternary rutile oxide $(Cr,Ta,Ti)_2O_4$ identified in R1 and R3 in Fig. 14(b). Additionally, the complex ternary $(Nb,Ta)_{10}O_{25}$ oxide, whose cation configuration is unknown, shows a wide range of formation energies depending on composition. Eighty-eight configurations of the cation site occupancy were enumerated and their formation energies calculated. From here, the lowest formation energy from the cation configurations whose Nb:Ta occupancy ratio was roughly 2:1 was chosen for inclusion in the oxide layer composition predictions (80 meV/atom above the convex hull at 0 K at a minimum). Table 5 indicates that there is a nonzero amount of Ti in this oxide, as well, which likely stabilizes it at high temperatures by increasing the configurational entropy.

3.5. Oxide layer predictions from oxygen & nitrogen grand potential minimization

Modeling the complex oxide layer structure formed during oxidation of these alloys via grand potential minimization results in a multi-layered, composite oxide and nitride scale, much like the observations detailed in Section 3.3. The oxide and nitride layer calculations for all three alloys is shown in Fig. 16; the phase fraction of each phase present at a given value of μ_N and μ_O, identified as "*μ-step*" in the figure, is shown for a range of chemical potential values. Phases which match experimental characterization are highlighted in each graph's legend. Each phase fraction graph should be thought of as a theoretical cross section of an oxidized alloy sample, with the base alloy on the left and the free surface of the oxide scale on the right. Each reaction zone identified in Section 3.3 roughly correlates with regions of the phase fraction graph, though many oxides and nitrides that appear are not identified experimentally. This is discussed in Section 4.

Table 6
Formation energy and minimum energies above the convex hull for all three alloys for each oxide simulated in this work. For energies above hull $<10^{-6}$ eV/atom, it is assumed they are on the convex hull and labeled "0.0 (on hull)".

Oxide	Formation energy (eV/atom)	NbCr – Minimum energy above hull (eV/atom)	NbCrTi – Minimum energy above hull (eV/atom)	NbCrTiTa – Minimum energy above hull (eV/atom)
NbO	−2.292	0.0 (on hull)	0.404	0.223
Cr_2O_3	−2.311	0.0 (on hull)	0.0 (on hull)	0.0 (on hull)
$NbCrO_4$	−2.654	0.346	0.346	0.346
TiO_2	−3.449	N/A	0.0 (on hull)	0.0 (on hull)
$NbTiO_4$	−3.176	N/A	0.091	0.091
$CrTiO_4$	−3.038	N/A	0.0 (on hull)	0.0 (on hull)
Cr_2TiO_5	−2.694	N/A	0.127	0.127
$CrTaO_4$	−2.876	N/A	N/A	0.0 (on hull)
$TaTiO_4$	−3.277	N/A	N/A	0.0 (on hull)
$(Nb, Ta)_{10}O_{25}$	−2.584	N/A	N/A	0.084

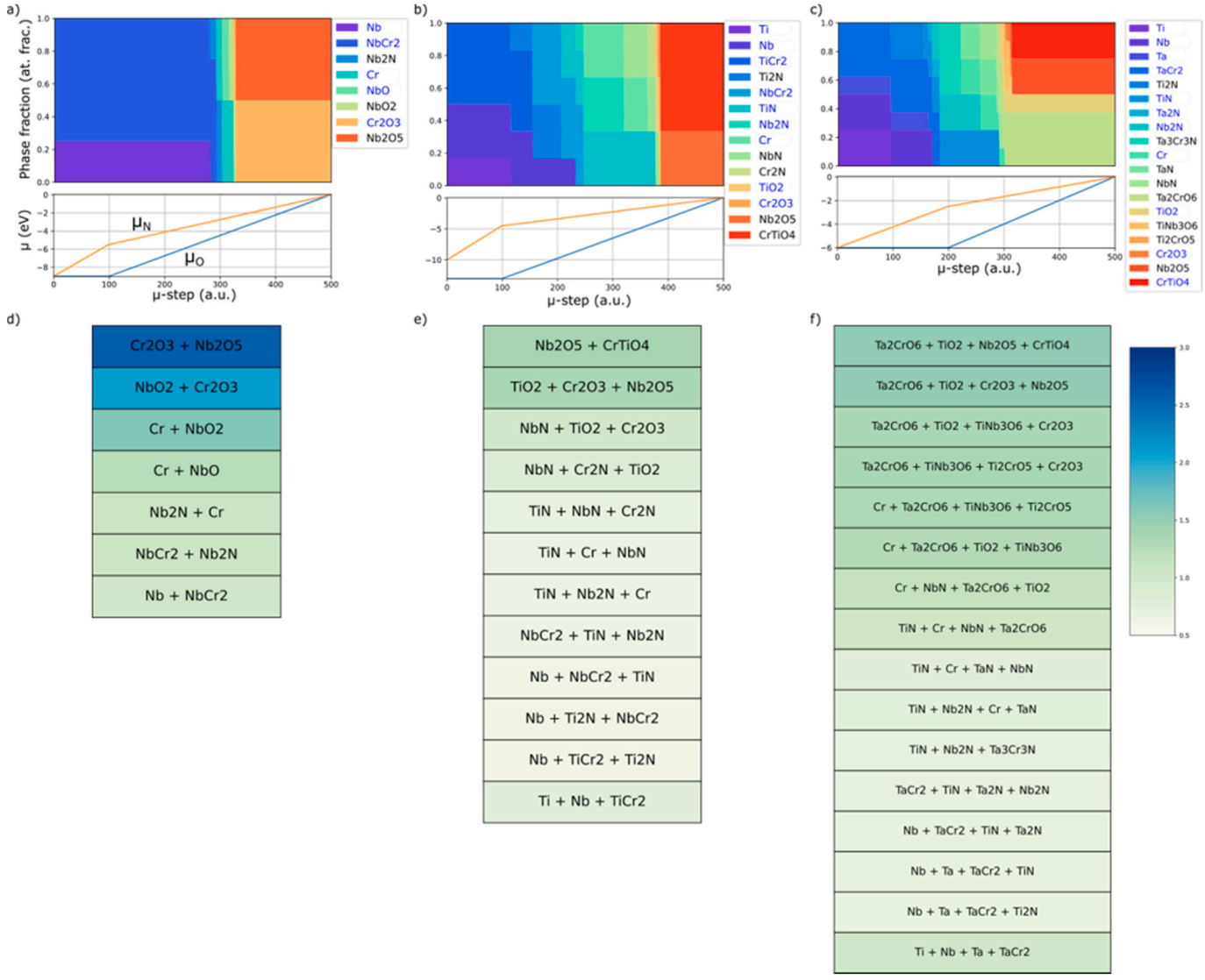


Fig. 16. Oxide layer predictions for (a) CrNb, (b) CrNbTi, and (c) CrNbTaTi alloys at 1200 °C. Phases identified experimentally are highlighted in blue. The phase fraction as a function of composite chemical potential μ for each alloy is shown, with the chemical potential for each alloy shown below. The weighted PBR of each layer, (d) CrNb, (e) CrNbTi, and (f) CrNbTaTi with the color scale for all three on the right, is shown relative to the weighted molar volume of the base alloy, is weighted by the phase fraction of each phase.

A common method for estimating oxide adherence strength is the Pilling-Bedworth ratio (PBR), which is the ratio of the normalized molar volume of the oxide to the metal substrate [37]. Generally, values between 1 and 2 indicate strong bonding between the oxide and substrate. PBR values below 1 indicate non-continuous scale formation and greater than 2 indicate compressive stresses that lead to scale cracking

and spallation. We extend this model to the complex, multi-phase oxide scales by calculating the weighted average molar volume of the alloy and the weighted average molar volume of each oxide/nitride layer and comparing the two. The results for all three alloys are shown layering color maps in Fig. 16(*d*–f). Each layer represents a distinct set of phases present in each alloy, with the color matching the phase fraction weighted PBR for the layer, with all three sets of layers sharing a common color scale, displayed in Fig. 16(d).

4. Discussion

4.1. Influence of compositional complexity on microstructure and performance

The alloys in this work, while microstructurally different, all formed a combination of BCC and Laves phases. As reported in other work [33] alloying additions of Ti and/or Ta to CrNb increase the phase fraction of BCC phase at the expense of Laves phase. This is not surprising, since it is well known that Ti and Ta are BCC stabilizers in refractory alloy systems, as described by Senkov et al. [1] and Butler et al. [33]. However, the interesting aspect here is that adding Ti in equiatomic concentration to CrNb (e.g. CrNbTi) caused the microstructure to invert the matrix and precipitate phase, Fig. 2. In the binary case, CrNb forms a Laves matrix with BCC precipitates, while CrNbTi forms a BCC matrix with Laves precipitates, Fig. 2. Additionally, the BCC phase in the ternary alloy remains extensively rich in Ti, which is predicted to help offset the expected deleterious oxidation behavior of Nb-enriched phases, Table 1. With the additional presence of Ti, the CrNbTi alloy also formed a capping layer of TiO_2. Further additions of Ta (e.g. CrNbTaTi) facilitate an increase in the fraction of BCC phase. However, in the quaternary case, the BCC phase becomes extensively enriched in both Nb and Ta, which is hypothesized to diminish oxidation resistance due to the poor elemental behaviors of Nb and Ta, Table 1 [36]. Other work has also alluded to diminished environmental resistance, especially concerning nitrogen, of Cr_2O_3 with Nb additions [11,12,18]. These concepts are consistent with the observed mass change, where CrNbTaTi exhibited the largest mass gains, while CrNb and CrNbTi offered similar, yet superior environmental resistances, Fig. 3. This similarity in environmental performance for CrNb and CrNbTi is interesting, since their inherent reaction microstructures were distinctly different, Figs. 4 and 5. This is promising from an alloy discovery point of view, as Butler et al. [33] reported a reasonable enhancement of room temperature ductility (compression) when Ti is added to CrNb.

4.2. Environmental reaction kinetics

All of the alloys investigated in this work formed some capacity of Cr_2O_3 after oxidation at 1200 °C in static air. This was in the form of dispersed regions for CrNbTaTi, while the CrNb and CrNbTi alloys both formed continuous, external Cr_2O_3 scales, Fig. 5. External Cr_2O_3 formation has also been reported during the high temperature oxidation of Cr(Nb) and Cr(Ta) BCC solid solutions, single phase Laves alloys (e.g. Cr_2Ta, Cr_2Nb), as well as, Cr(x)-Cr_2(X) (X = Nb, Ta) eutectic alloy systems [12,18]. While Cr_2O_3 is classically considered a protective oxide layer if formed in the appropriate morphology, it has the potential to facilitate the formation of volatile species (e.g. CrO_3) at elevated temperatures [41,42]. The anticipated linear rate loss due to the volatilization of CrO_3 has been quantified under various conditions on pure Cr_2O_3 [42] and during the oxidation of a Ni–Cr alloy [41]. Based on the temperatures and static conditions in this work, it is estimated that the linear rate loss due to the volatilization of Cr_2O_3 is on the order of 10^{-2} to 10^{-3} (mg/cm^2 h) [41,42]. Over the course of 24 h, this would account for ~0.024–0.24 mg/cm^2 total mass loss occurring in tandem with mass gains due to combined oxidation and nitridation, Fig. 3. One other complicating factor is the fact that testing was conducted in lab air, which contains a level of humidity. It is likely that chromium hydroxides are also forming to some extent and would increase the severity of volatilization. However, further work is needed to quantify the extent of hydroxide formation in these alloys. It is interesting to note that the Cr_2O_3 scale that formed on the CrNbTi alloy appeared to exhibit lower porosity compared to the outer Cr_2O_3 scale that formed on CrNb, as shown most prominently in the SEM-EDS maps in Fig. 5(a)–(b). It is possible that the additional outer TiO_2 scale that formed above the Cr_2O_3 layer in the CrNbTi alloy may aid in the retention of Cr_2O_3 for longer periods. Although, this outer TiO_2 scale did exhibit a great deal of porosity, Fig. 5(b). The CrNb and CrNbTi alloys both exhibited nearly parabolic oxidation kinetics throughout oxidation exposure.

The reaction kinetics for the CrNbTaTi alloy displayed two distinct regimes, Fig. 3(b). The first was parabolic ($n = 0.43$), extending out to 8 h of oxidation at 1200 °C, followed by a transition to linear kinetics ($n = 1.01$) out to 24 h. Considering the highly-repeating, multi-layered oxide structure observed, Fig. 11, it is likely that oxide scale cracking and/or spallation at longer exposure times is driving the shift in oxidation mechanism. It is clear that as the repeating oxide structure gets thicker, there is an unfavorable build-up of stress causing a combination of cracking between layers, Fig. 11(f), as well as through thickness cracks and the development of major voids, Fig. 11(e).

4.3. Oxide morphologies and mechanisms

CrNb formed a two-layered oxide structure, followed by a pronounced region of internal oxidation/nitridation, Fig. 5. The outer most oxide layer was determined to be Cr_2O_3, with an underlying layer of $CrNbO_4$, Fig. 7(b). This is similar to the work of Brady et al. [12] who reported the formation of an outer Cr_2O_3 scale on top of internally nitrided regions for Cr-rich, Cr(Nb) – Cr_2Nb alloys [12]. It is hypothesized that this outer Cr_2O_3 layer forms in CrNb due to the outward diffusion of Cr to the alloy surface. While the initial free surface interface could not be trivially identified after oxidation exposure, it is thought to exist near the Cr_2O_3 and $CrNbO_4$ boundary. This is supported by the presence of remnant BCC matrix phase interspersed in the lower portions of $CrNbO_4$, Fig. 7(b). In this manner, the local diffusion of Cr to the surface creates some level of Nb-enrichment below the Cr_2O_3 layer. This likely initiates the local formation of Nb_2O_5 and selective reaction with Cr_2O_3 to form $CrNbO_4$, albeit slightly enriched in Nb, Table 3. This intermediate $CrNbO_4$ layer appears to play a significant role in oxide adherence, as spallation of the outer oxide scale was not observed for the CrNb alloy in this work (or the CrNbTi alloy), but has been reported to be significant for other Cr-rich, Cr(Nb)–Cr_2Nb based alloys, leading to substantial mass loss kinetics [12,18].

Below the outer oxides of the CrNb alloy, it is clear that both nitrogen and oxygen react with a significant portion of the underlying alloy, Figs. 5(a) and Fig. 7(c). In particular, the center cores of the BCC matrix formed NbO, while the remaining extent of oxygen was observed to segregate to Laves phase regions, Fig. 7(c). The formation of NbO is expected due to the high concentration of Nb and reduced partial pressures of oxygen present in these regions. This is combined with the high thermodynamic stability of the monoxide compared to the pentoxide. It was also observed that surrounding BCC regions were decorated with Cr_3Nb_3N necklace structures, Fig. 7(c). This complex oxide has also been observed in related Cr-rich eutectic alloys [11,12,18]. The regions surrounding the prior BCC areas retained the Laves phase structure, but were found to be enriched in oxygen.

Compared with the work of Brady et al. [11,12,18] on Cr-rich, CrNb alloys, it is clear than the increased concentration of Nb in this work drives a more complex oxide scale and internal reaction product. For example, Brady et al. report the external formation of Cr_2O_3, followed by an internal layer of nitrides, whereby the Cr(Nb) solid solution and Cr_2Nb Laves phase are equally nitrided. Below this layer, only the Cr_2Nb Laves phase is nitrided, while the Cr(Nb) solid solution phase remains metallic. Based on the results of this work, increased Nb concentrations facilitate the formation of an intermediate $CrNbO_4$ layer and complicate

the internal reaction product, as described above. TEM results suggest that both the BCC and Laves phases contribute to nitridation in the CrNb alloy. This shift is likely due to the increased fraction of Nb in the BCC phase, compared to the work of Brady et al. [11,12,18], along with the higher thermodynamic tendency (more negative) for the formation of Nb-based nitrides versus Cr-based nitrides [43]. This is enhanced further with subsequent additions of both Ti and Ta, which both exhibit high thermodynamic tendencies for formation compared to Cr, with TiN being the most negative.

The addition of Ti to CrNb promoted a significant microstructural change compared to the binary alloy. The CrNbTi alloy was found to consist of a BCC matrix with Laves phase precipitates, Fig. 2(e). The oxide scale of the ternary alloy was more complex and consisted of an outer layer of TiO_2, followed by Cr_2O_3 and underlying $NbTiO_4$, Figs. 5(b) and Fig. 9. However, this shift in oxide species in CrNb versus CrNbTi (e.g., $CrNbO_4$ vs $NbTiO_4$) seemed to have little influence on the overall environmental performance of the alloys. In addition, an Nb_2TiO_7 phase was observed between the outer TiO_2 and Cr_2O_3 layers, Fig. 9(b). Underneath the outer oxide scales, a highly complex distribution of oxides and nitrides were observed, Fig. 10(b). In particular, the prior BCC regions were pervasively decorated with Cr_3Nb_3N, TiO_2 and small pockets of Nb_2N. The prior Laves phase regions similarly form a high distribution of Cr_3Nb_3N and TiO_2, along with Cr_2O_3. The most internal regions of the oxide scale consisted of remnant BCC and Laves phases, along with a distribution of highly faceted TiN particles throughout the BCC regions, Fig. 11.

The CrNbTaTi alloy displayed vastly different reaction product morphologies and inferior environmental performance, as compared to CrNb and CrNbTi. In particular, the quaternary alloy formed a repeating, discontinuous structure that was plagued with cracks and suffered from extensive oxide spallation, Fig. 12. Simpler, alternating structures have been reported for Cr-rich CrTa and CrNb based alloys [11,12,18]. The outer oxide layers consisted of a variety of mixed, rutile-structured oxides, along with several other distinct complex oxides (e.g., Cr_2TiO_5, $(Nb,Ta)_{10}O_{25}$), depending on the oxidation depth. The mixed, rutile-structured oxides exhibited only subtle variations in local composition, Fig. 13(c) and Table 6. This demonstrates that increasing compositional complexity appears to promote a level of "confusion" with regard to oxide formation. This also correlates well with the expected level of competition between oxides, as predicted in the modeling results for the CrNbTaTi alloy. However, the internal nitride structures of the CrNbTaTi alloy were similar to the CrNbTi alloy, although much finer in scale due to the feature size in the starting microstructure, Figs. 2 and 15. This is likely due to the similar stabilities of Ta and Nb based nitrides, paired with the ability for Ta and Nb to substitute for one another based on their periodic grouping [43].

4.4. Comparison between grand potential calculations and experiment

The grand potential calculations of phase equilibrium for all three alloys agree well with experiment, with some notable discrepancies to be discussed. Introducing nitrogen as a potential oxidant into the system does not necessarily preclude the formation of nitrides below the oxide scale, according to these calculations. For instance, if $\mu_N = \mu_O$, oxides will predominantly form, consuming all available metal cations and preventing nitride formation. Specifying the chemical potential of nitrogen and oxygen independently to better align with the experimental results, however, can give insight into the diffusion of gaseous species into the alloy. If, instead, $\mu_N > \mu_O$, then nitrides will form at low values of μ_O, which agrees with the analysis in Section 3.3. We can conclude that since a higher chemical potential of nitrogen relative to oxygen corresponds well to the experimental results, nitrogen likely diffuses more readily through the oxide scale and into the metal lattice.

In the case of CrNb, the main difference between experiment and theory stems from the formation of high temperature complex oxides and nitrides (e.g., Cr_3Nb_3N and $CrNbO_4$), Fig. 16(a). Grand potential calculations predict the component phases which make up these two phases – $NbCr_2$, Nb_2N, and Cr for Cr_3Nb_3N and Cr_2O_3 and Nb_2O_5 for $CrNbO_4$. Calculations of the formation energy of $CrNbO_4$ and Cr_3Nb_3N show they are 0.35 eV/atom and 0.018 eV/atom above the convex hull for this system, respectively, which are energy barriers surmountable by temperature effects.

The ternary alloy CrNbTi exhibits a similar pattern to CrNb, as shown in Fig. 16(b). The internal reaction zone in Fig. 11 matches well with the phases predicted for low μ_N and μ_O, but the composition does not match the IRZ in Fig. 10; where the grand potential predicts the formation of Ti_2N, TiN, Nb_2N, and elemental Cr, experiment shows a combination of TiO_2, Cr_3Nb_3N, Nb_2N, and Cr_2O_3. This matches relatively well, and temperature stabilization effects may explain some of these discrepancies; it is also likely that the concentrations of N and O within the alloy and scale are higher than the corresponding μ_N and μ_O shown in Fig. 16; without reliable measurements of solubility of gaseous N_2 and O_2 in each compound, it is difficult to predict the concentration of each species in the scale during experiment. Grand potential calculations also predict a different complex ternary oxide from the one shown experimentally in the external oxide scale; instead of TiO_2, Cr_2O_3, Nb_2N, and the mixed rutile $NbTiO_4$, Fig. 16(b) shows a combination of Cr_2N, TiO_2, Cr_2O_3, Nb_2O_5, and the mixed rutile $CrTiO_4$ for μ_N and μ_O near to 0.

The oxide scale formed on CrNbTaTi appears to form cyclically, with repeated layers of oxide phases. The internal oxidation front in Fig. 15 matches well with predictions of the low chemical potential, matching Laves, BCC, and TiN (grand potential calculations also predict Ti_2N, though this was not identified). Within the IRZ in Fig. 14(c), where the Laves phase and $(Nb,Ta)_2N$ form with Cr_2TiO_5 and $TaTiO_4$, grand potential calculations show that there is significant formation competition amongst many ternary oxides and nitrides. At the oxide scale in Fig. 14 (b), this competition increases, with many different oxides soluble within each other forming simultaneously, including TiO_2, Ta_2CrO_6, $TiNb_3O_6$, Ti_2CrO_5, Cr_2O_3, Nb_2O_5, and $CrTiO_4$. The absence of $(Nb,Ta)_{10}O_{25}$ from the predicted composition is not surprising, as Table 6 shows it to be 80 meV/atom above the convex hull. Considering that the lack of data on composition and cation configuration, the structure and energy used in this study is likely not the equilibrium configuration, either at 0 K or at finite temperatures. Interestingly, $CrTaO_4$, $CrTiO_4$ and $TaTiO_4$ are all predicted to be on the convex hull, and appear in the mixed $(Cr,Ta,Ti)_2O_4$ identified in the oxide in Table 5, but only $CrTiO_4$ appears in the grand potential calculations. Here, then, the grand potential calculations also predict significant formation competition, which agrees experimentally. Because thermodynamics cannot predict cracking or porosity during formation of the oxide scale, the cyclical formation of alternating Cr, Ti, and Nb/Ta-rich oxide regions is not captured here.

One potential explanation for the poor oxidation performance and cyclically forming oxides of the quaternary CrNbTaTi relative to the binary CrNb and ternary CrNbTi alloys is the adherence of the oxide scale to the metal substrate, as measured by the weighted PBR calculations in Fig. 16(*d*–f). For CrNb, phases predicted by the oxygen/nitrogen grand potentials show very high weighted PBR values near the exterior and ideal PBR values in the IRZ, indicating potentially excessive compressive stress in the oxide, leading to cracking or porosity (Fig. 16 (d)). This still holds true when substituting the predicted Nb_2O_5 (PBR = 2.65) with the observed $NbCrO_4$ (PBR = 2.30). This could explain the observed porosity in the oxide scale in Fig. 5(a), which is likely a manifestation of both volatility and PBR effects. The opposite trend appears in CrNbTi (Fig. 16(e)) with the influence of low molar volume nitrides resulting in a weighted PBR <1, which agrees with the internal porosity visible within the first IRZ identified in Fig. 5(b). The weighted PBR estimates of CrNbTaTi show a similar potential, with many of the complex layers in the external scale with ~1 < PBR < ~1.5. However, the presence of low molar volume nitrides in the IRZs could lead to porosity, cracking, and oxygen ingress. Since the phases predicted by the grand potential calculations do not agree well with the CrNbTaTi alloy,

these results, at best, suggest a possible cause for the cyclical oxidation of the sample rather than present a strong candidate for it.

4.5. Design considerations

The results presented here demonstrate that there is available potential for optimizing environmental resistance in this alloy space and that there is potential for short use consideration of Nb(Cr)–Cr_2Nb based alloys. A common view for Nb or Ta rich BCC + Laves forming refractory alloys is that suitable oxidation resistance is generally only observed in alloys with high Laves phase content, due to the large reservoirs of Cr and the anticipated formation of Cr_2O_3. This is problematic because Laves phase, while providing strength, can also induce a drastic reduction in ductility. However, the similar performance of the CrNb and CrNbTi alloys in this work demonstrates that comparable environmental resistance can also be achieved in an alloy with a predominant fraction of Cr-lean BCC phase, depending on the particular alloying additions (e.g. Ti). Butler et al. [33] also demonstrated that additions of Ti to CrNb and Ta to CrNbTi improved room temperature ductility, albeit, with some loss in high temperature strength. In addition, the concentrated nature of the alloys promotes the formation of complex oxide species that offer sluggish oxidation kinetics appropriate for short-use and/or expendable environments. These oxides were particularly useful when considering the oxide spallation characteristics of Cr(Nb)–Cr_2Nb alloys versus Nb(Cr)–Cr_2Nb. In addition to the compositional modifications investigated in this work, it may also possible to optimize properties further through microstructural tailoring, whether by conventional thermo-mechanical processing (TMP) or considering near-net-shape manufacturing approaches (e.g. additive manufacturing, powder metallurgy).

5. Conclusions

This work reports the microstructures and oxidation behaviors (1200 °C) for CrNb, CrNbTi, and CrNbTaTi. The following key points can be made:

1) CrNb, CrNbTi, and CrNbTaTi all form BCC + Laves (C15) microstructures. The addition of Ti to CrNb inverts the microstructure to predominantly a BCC matrix with Laves (C15) precipitates. Surprisingly, this BCC matrix in the ternary alloy remains extensively rich in Ti. The addition of Ta to the ternary increases the overall phase fraction of BCC matrix, but also retains Laves (C15) precipitates.
2) The binary and ternary alloys both displayed nearly parabolic, single regimes of environmental reaction, with no evidence of oxide spallation. Comparison with prior work suggests that the formation of complex oxide species (e.g. $CrNbO_4$ and $CrTiO_4$) may promote extended scale adhesion. Conversely, the CrNbTaTi alloy displayed nearly parabolic kinetics out to 8hrs, followed by a transition to linear kinetics. This was ultimately attributed to the extensive cracking and oxide spallation throughout oxidation exposure.
3) The CrNb and CrNbTi alloys both formed distinct oxide layers atop a series of internally reacted zones at 1200 °C. In particular, the CrNb alloy formed an outer layer of Cr_2O_3 with an underlying layer of $CrNbO_4$ and interspersed, remnant BCC metal regions. This was followed by an internally reacted zone (IRZ) consisting of NbO, Cr_3Nb_3N and remnant Laves phase. Similarly, the CrNbTi alloy formed an outer layer of TiO_2, followed by Cr_2O_3 and underlying regions consisting of $NbTiO_4$ with interspersed Nb_2N. The internal regions of the CrNbTi alloy were more complex, consisting of TiO_2, Cr_2O_3, Nb_2N, Cr_3Nb_3N, and a combination of remnant BCC matrix, Laves phase and TiN deeper into the alloy.
4) The oxide product of the CrNbTaTi alloy was the most complex and consisted of a discontinuous, repeating structure with a high density of cracks and voids. The oxide layers were found to consist of a combination of mixed, rutile structured oxides, along with a few distinct oxides (e.g., Cr_2TiO_5, $(Nb,Ta)_{10}O_{25}$), depending on depth into the alloy.
5) Modeling the identified oxides in this work using DFT, it is found that even at 0 K, many of the oxides, including the mixed rutile oxides, are either the stable oxide or are close enough to the convex hull that vibrational energy and entropy associated with temperature will make them the stable oxides at 1200 °C. Oxygen vacancy formation energy calculations show that the complex ternary rutile oxides are all promising candidates for oxidation protection. The exception is $NbCrO_4$, which contradicts previous research and requires further study.
6) Predictions of the oxide scale composition from thermodynamic grand potential calculations show that the ab initio data agrees well with CrNb and CrNbTi oxidation experiments, while the CrNbTaTi predictions indicate significant formation competition amongst many different ternary oxides. Calculations of the weighted PBR for each oxide layer show that the observed nitrides might be the cause of the oxide scale porosity, while many of the ternary oxides show strong potential adherence.

Credit author statement

Todd Butler: Conceptualization, Methodology, Formal Analysis, Investigation, Data Curation; Supervision, Writing – Original Draft, Review & Editing; **Oleg Senkov:** Methodology, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Review & Editing; **Tinuade Daboiku:** Formal Analysis, Writing – Review & Editing; **Mike Velez:** Investigation, Data Curation, Writing – Review & Editing; **Haydn Schroader:** Investigation, Data Curation, Writing – Original Draft, Review & Editing, Software and Modeling; **Logan Ware:** Investigation, Data Curation, Writing – Original Draft, Review & Editing, Software and Modeling; **Michael Titus:** Investigation, Data Curation, Writing – Original Draft, Review & Editing, Software and Modeling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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